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Export Risk Assessment in the Veneto Manufacturing Sector

A Network-Gravity Machine Learning Approach

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INTRODUCTION

The structural core of the Veneto region is rooted in the Marshallian industrial district which is a localized cluster of small and medium sized enterprises (SMEs) that operate in complementary industries, benefiting from external economies of scale and scope that let them to reduce average unit costs and facilitate joint production of goods. The geographical proximity is a crucial characteristic of these specialized firms that can share know-how, resources and infrastructures to improve the aggregate efficiency of the industrial district. This concept was fundamentally redefined by Becattini (2017) who argued that the industrial district is not merely a geographic concentration of production, but a social and economic environment that fosters unique synergy and where the social community and the firm population are inextricably linked.

Central to this genesis is the technical division of labour among a dense population of specialized firms. Unlike the Fordist model of vertical integration, the Marshallian district operates through vertical disintegration. In this configuration, the production of a final good is decomposed into a series of discrete, sequential phases. Rather than one large factory overseeing the entire lifecycle of a finished product, the district hosts a multitude of independent units. For example, one firm may specialize exclusively in the design and prototyping phase, another in the forging of raw components, a third in specialized heat treatments or finishing, and a final entity in assembly and branding. This fragmentation allows small firms to achieve a high degree of technical efficiency within their specific niche. The "industrial atmosphere" described by Becattini (1990) ensures that knowledge, skills, and innovations circulate freely between these phase-specialized actors, creating a collective efficiency that rivals the internal economies of large-scale enterprises. This localized thickness has fostered a diverse array of industrial districts that serve as the regional pillars of international trade. In Belluno, there is the eyewear ecosystem that represents a global hegemony in optical manufacturing. Conegliano and Valdobbiadene (located in Treviso) are agro-industrial areas that have successfully transitioned from a localized agricultural commodity to a globally recognized premium brand as that of Prosecco. The footwear of the Brenta Riviera (through Venice and Padua) is characterized by a high-prestige hub with world's most prominent fashion houses specializing in luxury footwear. Vicenza, standing as one of Italy's leading precious metal

clusters, leads the way in the goldsmith production and is a primary contributor to the high unit values of the region's export data. The last province is also known for leather processing, utilizing advanced chemical and mechanical phases to produce advanced textiles. Treviso is a versatile district focused on polymer processing, providing essential technical components for a variety of domestic and international industrial supply chains. Finally, Verona is an engineering vanguard cantered on the production of sophisticated heating and air conditioning systems. The structural persistence of these districts is predicated on the geographic concentration of expertise and human capital. In these clusters, the "industrial atmosphere" is not a mere metaphor but a measurable economic advantage, where the proximity of specialized suppliers reduces transaction costs and accelerates the diffusion of innovation. This regional configuration provides the essential micro-level context for understanding the "superstar exporters" phenomena explored in later chapters; the elite exporters analysed in this study do not emerge in a vacuum but are the lead firms nurtured by this highly specialized, district-based productive fabric.

As the global economic landscape shifted toward increased modularity and international competition, the traditional, self-contained district underwent a structural metamorphosis. Corò and Grandinetti (1999) identify these evolutionary patterns as a transition from local thickness to global connectivity. Historically, the district relied on idiosyncratic local knowledge—tacit skills passed down within the community. However, the modern iter of the district requires a shift toward open networks. The evolution is marked by the rise of lead firms that often take on the role of system integrators. While they continue to rely on the local population of phase-specialized firms for production, they manage the global pipelines that connect local manufacturing excellence to international markets. The transition involves moving beyond the territorial boundary: firms no longer compete solely based on local synergies but on their ability to integrate into Global Value Chains (GVCs). As observed by Corò and Grandinetti, the resilience of the Veneto model now depends on the ability of these specialized firms to serve multiple global hubs, effectively decoupling their phase-specific expertise from a single local final-good producer. In this evolved state, the population of firms retains its phase-based expertise but gains the agility to operate within high-tech and medium-tech GVCs. Entering a GVC exposes firms to international quality standards and the specific needs and wants of the customers in each geographic area the firm has diversified its exports.

This structural openness is what allows the Veneto vanguard to sustain high unit values even in the face of macro-economic volatility. In the econometric context, export value is a fundamental metric representing a firm's successfully realized market share and its qualitative pricing power. Export value represents the total monetary return on a firm's international activity. It is a composite indicator of efficiency (the ability to minimize production costs relative to global competitors) and upgrading (the transition toward higher-added-value products). Integration into GVCs fundamentally alters firm growth trajectories, leading to higher total export values—specifically by increasing their domestic value-added (DVA) or managing foreign-origin added value (FVA).

Contemporary international trade architectures are increasingly defined by profound firm-level heterogeneity, a phenomenon wherein a marginal cohort of "superstar exporters" accounts for a disproportionate share of aggregate national trade volume and value. For export-oriented economies such as Italy, understanding the structural dynamics and relational intricacies of these trade networks is critical for both macroeconomic forecasting and the formulation of strategic industrial policy. While traditional gravity models of trade, first introduced by Walter Isard in 1954, effectively predict aggregate bilateral flows based on the interplay of economic mass and geodetic distance¹, they often lack the granularity required to capture the nuanced topological positioning of individual firms within Global Value Chains. Furthermore, conventional linear frameworks frequently fail to isolate the compounding effects of institutional frictions—specifically political and credit risks—within destination markets. This research bridges this gap by applying bipartite network analysis with mixed-effects econometric modelling and machine learning techniques. By disaggregating trade flows to the firm-country level, this study isolates the specific premiums associated with physical volume, geographical proximity, institutional stability, and network centrality. The primary objective is to investigate the structural determinants of export monetary value and pricing power—proxied via export unit values—within the manufacturing ecosystem of the Veneto region. By integrating the classical Gravity Model with the Alchian-Allen "spatial quality sorting" conjecture, firm-level network topology, and

¹ The gravity model of trade predicts that bilateral trade volume between two countries is directly proportional to their economic size (GDP) and inversely proportional to the distance between them. Larger economies produce more and consume more, leading to higher trade volumes (positive relationship); while increased geographical distance raises transportation and transaction costs, reducing trade volumes (negative relationship).

institutional risk analysis, this study aims to identify the fundamental drivers of high-value-added exports. Through this multi-dimensional approach, we aim to map the structural archetypes of the Veneto manufacturing sector, providing a rigorous assessment of how relational prestige and institutional risk-mitigation strategies dictate the competitive positioning of firms in the global marketplace.

CHAPTER I

THEORETICAL FOUNDATIONS AND METHODOLOGICAL FRAMEWORK

1.1 Related Literature

First, this study addresses the comprehensive mapping of regional performance: what are the structural, topological, and institutional determinants of export value in the Veneto manufacturing sector? The theoretical foundation of this research lies at the intersection of internationalization risk, firm-level heterogeneity, and network topology. By synthesizing these domains, this research contextualizes the structural determinants of bilateral trade flows within the modern gravity model framework. As noted by Shepherd et al. (2019), the gravity equation remains the "workhorse" of international economics, providing a theoretically consistent method for estimating the impact of trade barriers and facilitators. The navigation of macroeconomic and institutional uncertainty is a central paradigm in international business literature. Eduardsen and Marinova (2020) provide a comprehensive integrative framework, demonstrating that risk is not merely an exogenous barrier but a dynamic determinant that shapes the scale, scope, and mode of firm internationalization. Empirical evidence consistently shows that institutional fragility heavily penalizes trade flows. Specifically, Agarwal and Feils (2007) demonstrate that political risk acts as a severe friction for Canadian-based export and Foreign Direct Investment (FDI) firms, structurally deterring market entry and depressing trade volumes. Söderlund and Tingvall (2014) further refine this by illustrating the dynamic effects of institutions, noting that the long-term stability of a destination's legal and political framework is a prerequisite for sustained firm-level export growth.

Complementing the impact of political instability, the role of financial frictions and institutional quality is equally critical. Rienstra-Munnicha and Turvey (2002) highlight the direct relationship between credit risk and export viability, illustrating how default probabilities in destination markets contract trade unless mitigated by mechanisms such as export credit guarantees. Demir and Hu (2020) argue that destination institutions significantly shape exporter dynamics, where poor institutional quality acts as a barrier that only the most resilient firms can bypass. Furthermore, Kraus, Ambos, Eggers, and Cesinger (2015) bridge the gap between geographic distance and risk, finding that the

physical and psychic distance to a destination market profoundly amplifies managerial perceptions of risk, thereby altering internationalization decisions. Collectively, this literature validates the necessity of compressing highly collinear political and credit risk indicators into a singular, orthogonal index to accurately measure destination market friction. Traditional macroeconomic trade theories relied on the assumption of a representative firm. However, the introduction of firm-level heterogeneity by Melitz (2003) revolutionized the field, proving that only the most productive firms can overcome the sunk costs of entering foreign markets. This theoretical breakthrough paved the way for the empirical identification of "superstar exporters"—a phenomenon where a fractional percentage of hyper-productive firms dominates national export aggregates (Freund & Pierola, 2015). Also, Farole and Winkler (2011) emphasize that firm location and regional clustering—characteristics inherent to Veneto's industrial districts—further amplify these heterogeneous advantages by providing localized spillover effects. The severe right-skewness and Pareto-like distribution observed in the export monetary value and physical trade volume data directly corroborate these foundational models.

Subsequently, the research explores the economic dividends of relational capital: to what extent does the position within the global trade network act as a predictor of the firm's foreign pricing power? Beyond baseline productivity, modern literature emphasizes the topological positioning of firms within the Global Supply Chains. Chaney (2014) highlights the network structure of international trade, positing that firms do not merely export to isolated markets but utilize existing counterparty networks to incrementally search for new buyers. Consequently, network centrality metrics—such as *out-degree* (extensive margin) and *eigenvector centrality* (network prestige)—serve as vital proxies for a firm's integration into core global hubs (De Benedictis & Tajoli, 2011). The spatial decay of physical trade volume relative to monetary value is theoretically grounded in the Alchian-Allen conjecture. As empirically confirmed by Hummels and Skiba (2004), per-unit transportation costs disproportionately penalize low-margin bulk goods over long distances, effectively "shipping the good apples out." The literature perfectly contextualizes the volume-value asymmetry observed in distant, high-value Veneto export destinations such as the United States justifying the integration of geographical distance and unit-value composition into the mixed-effects framework. This study probes a significant theoretical tension in contemporary trade geography: how does institutional

risk distort the Alchian-Allen quality sorting effect in the Veneto manufacturing sector? While Hummels and Skiba (2004) confirm that distance filters for quality, Demir and Hu (2022) demonstrate that institutional similarity and quality are equally decisive in determining export sophistication. This suggests that pricing power in Veneto is not just a function of distance-induced quality sorting, but of the institutional alignment between the region and its trading partners.

Finally, this research addresses the methodological limitations of traditional linear gravity models. While Linear Mixed Models (LMM) are essential for decomposing variance and handling unobserved firm-country heterogeneity, they often fail to capture the compounding effects of trade barriers. Recent scholarship in computational economics (Varian, 2014; Mullainathan & Spiess, 2017) advocates for the integration of machine learning to handle high-dimensional, non-linear interactions. Specifically, Verstyuk and Douglas (2022) argue that "machine learning the gravity equation" allows for the discovery of complex, non-linear patterns that traditional econometrics miss. By utilizing Extreme Gradient Boosting (XGBoost), this research acknowledges that the impact of risk may not be additive but multiplicative when combined other predictors such as firm's prestige and connectedness in the trade network, providing a more robust predictive assessment of Veneto's export performance in the volatile 2024 global economy.

1.2 Methods, Materials, and Data

The empirical analysis was conducted using R software and utilized a highly granular, cross-sectional dataset of bilateral export flows from the Veneto region for the 2024 fiscal year. The primary trade microdata, capturing the physical quantity (kg) and monetary value (€) at the firm-destination level, were sourced from the Italian National Institute of Statistics (ISTAT). To account for macroeconomic scale and spatial friction, these flows were merged with country-level GDP data retrieved via the World Bank's World Development Indicators (WDI) and geodetic distances computed from the Veneto economic centroid using the geosphere R package. Finally, country risk profiles—encompassing both credit risks (Sovereign, Banking, and Corporate) and political risks (Expropriation, Transfer, and Political Violence)—were integrated from SACE (the Italian

Export Credit Agency). Rigorous cleaning protocols were implemented to ensure the integrity of the multi-source dataset. Complete-case filtering was applied to resolve missing values generated during the merging process, ensuring only perfectly matched bilateral observations were retained. To maintain internal consistency, GDP data originally denominated in USD were converted to EUR using the 2024 average exchange rate of 1.08 EUR/USD. To mitigate the extreme right-skewness characteristic of trade data, logarithmic transformations were applied to monetary value and quantity variables. Notably, extreme upper-tail outliers were preserved and managed through non-linear transformations during the feature engineering phase to prevent significant information loss; these outliers represent ultra-high-value "Made in Italy" shipments in sectors based on luxury goods such as *Manufacture of jewellery and goldsmith's articles in precious metals or coated with precious metals (Ateco 2007: 32.12.1)* and *Processing of precious and semi-precious stones for jewellery and industrial use (Ateco 2007: 32.12.2)*².

The data were structured as a directed bipartite network, mapping connections from Veneto's firms (source nodes) to foreign destination markets (target nodes). From this topology, we extracted three centrality measures:

- *Out_Degree*: a measure of the extensive margin (number of unique destination markets reached);
- *Out_Strength*: the total historical scale/volume of the exporting firm;
- *Eigenvector centrality*: a proxy for network prestige and integration into core global hubs.

To address severe multicollinearity among the six SACE risk metrics, Principal Component Analysis (PCA) was applied. The first principal component was extracted to serve as a standardized, orthogonal Institutional Risk Index, capturing roughly 91% of the latent variance of institutional friction. The target variable was operationalized as monetary value of exports (€) or as the unit value of exports (€/kg), where the latter serves as a theoretically grounded proxy for product quality and pricing power. A Linear Mixed Model (LMM) was estimated using maximum likelihood to account for the crossed,

² For the fiscal year 2024, Istat microdata identifies specific industrial niches with exceptionally high pricing power. Sub-sectors including Ateco code 32.12.1 and Ateco code 32.12.2 exhibit average export unit values of 4,063.72 €/kg and 6,103.37 €/kg, respectively. In the context of this study, these observations are retained as legitimate representations of the "superstar exporter" phenomenon; their extreme variance is addressed via logarithmic normalization and the non-linear partitioning capabilities of the XGBoost algorithm rather than through outlier exclusion.

hierarchical nature of the dataset; the model incorporates random intercepts for both the exporting firm u_i and the destination country v_j to calculate and adjust for intra-class correlation (ICC). The formal specification is defined as:

$$\begin{aligned} \ln(\text{Value}_{ij}) = & \beta_0 + \beta_1 \text{Risk}_j + \beta_2 \ln(\text{Distance}_{ij}) + \beta_3 \ln(\text{GDP}_j) + \beta_4 \text{Out_Degree}_i \\ & + \beta_5 \text{Out_Strength}_i + \beta_6 \text{Eigen}_i + u_i + v_j + \epsilon_{ij} \end{aligned} \quad (1)$$

where i denotes the exporting firm located within the Veneto region, while j denotes the specific foreign destination country receiving the export flow. $\ln(\text{Value}_{ij})$ represents the natural logarithm of the absolute monetary export value (or export unit value, depending on the model specification) recorded in 2024 for transaction flows moving from firm i to destination country j . The log-transformation ensures that the estimated β coefficients capture constant elasticities or semi-elasticities, while stabilizing sample variance. In the log-linear configuration, coefficients matching log-transformed predictors (β_2, β_3) represent elasticities, while coefficients attached to linear or standardized predictors ($\beta_1, \beta_4, \beta_5, \beta_6$) represent semi-elasticities, indicating the percentage change in export value driven by a one-unit increase in the covariate. Instead, β_0 represents the baseline intercept of the model, capturing the expected mean of the log-export value when all continuous and standardized covariates are held at zero. Risk_j measures the country-level Institutional Risk Index for nation j . $\ln(\text{Distance}_{ij})$ is the natural logarithm of the geodetic distance between firm i and target market j , mapping out the standard spatial friction parameter of the gravity equation. $\ln(\text{GDP}_j)$ is the natural logarithm of the destination country's Gross Domestic Product, controlling for the economic size and absolute demand potential of target market j . Out_Degree_i tracks the extensive margin of firm i 's export portfolio, capturing the total count of active foreign destination markets served by the enterprise. Out_Strength_i measures the overall intensity or scale of firm i 's total out-of-region trade mass across its complete export grid. Eigen_i represents the standardized eigenvector centrality metric for firm i , operating as the primary topological proxy for structural network prestige and supply-chain positioning. Furthermore, u_i captures unobserved firm-specific random effects, absorbing idiosyncratic enterprise-level traits like productivity, managerial capacity, and baseline scale. While v_j denotes destination country-specific effects, capturing unobserved macro-level traits,

institutional features, or shared demand conditions in nation j . Finally, ϵ_{ij} represents the idiosyncratic stochastic error term, assuming independent and identically distributed (i.i.d.) properties with a mean of zero, capturing remaining white noise across individual trade lines. The Equation (1) uniquely bridges microeconomic firm behaviours with macroeconomic spatial frictions, providing a rigorous framework for identifying the structural drivers of high-value-added exports.

1.2.1 Linear Mixed Models (LMM): Coefficient-Based Estimation

The LMM follows a parametric approach where the relationship between predictors and the response is modelled as a linear combination of fixed and random effects. The general form is:

$$y = X\beta + Zu + \epsilon \quad (2)$$

where $X\beta$ represents the fixed effects (coefficients): their estimation relies on the magnitude of X which means that if X is scaled by a factor of 1,000, the corresponding β must shrink by 1,000 to maintain the equality; this makes the model scale-variant. Zu represents random effects (e.g., firm-specific or country-specific intercepts), capturing unobserved idiosyncratic heterogeneity and serial correlation (e.g., some firms are just inherently better at exporting, and some countries are inherently harder to export to, regardless of the risk indicators). To facilitate model selection and nested model comparison via Likelihood Ratio Tests (LRT), models were estimated using Maximum Likelihood (ML) by setting REML = FALSE. This ensures that the deviance statistics are comparable across specifications with varying fixed-effect structures.

1.2.2 XGBoost: Rank-Based Split Optimization

XGBoost is a non-parametric ensemble of decision trees. It does not calculate global coefficients; instead, it iteratively adds functions f_k to minimize a regularized objective function:

$$\mathcal{L}(\phi) = \sum_{i=1}^n l(\hat{y}_i, y_i) + \sum_{k=1}^K \Omega(f_k) \quad (3)$$

where $\mathcal{L}(\phi)$ represents the total objective loss function to be minimized during training. The parameter argument (ϕ) represents the entire ensemble parameter space, encompassing all structural parameters across the collection of trained decision trees. $\sum_{i=1}^n l(\hat{y}_i, y_i)$ indicates that the training loss is summed across all n observations in the sample matrix. The loss function evaluates the distance between predictions and actual observations. It measures how well the model fits the historical training observations. $\sum_{k=1}^K \Omega(f_k)$ sums the structural complexity penalties across all K individual trees built within the sequential gradient boosting chain. The regularization term penalizes overly complex models to ensure generalizability.

To find the optimal split at a node, XGBoost searches for the feature j and threshold t that maximizes the Information Gain:

$$Gain = \frac{1}{2} \left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda} \right] - \gamma \quad (4)$$

where G_L is the sum of gradients (first derivative) for all observations falling into the proposed Left child node, while G_R is the sum of gradients for all observations falling into the proposed Right child node. H_L acts as the sum of Hessians (second derivative) in the Left child node, while H_R acts as the sum of Hessians in the Right child node. $\frac{G_L^2}{H_L + \lambda}$ is the mathematical score of the Left child node if the split is executed. A higher value indicates that grouping these specific observations significantly cleans up and isolates a coherent cluster of directional errors. $\frac{G_R^2}{H_R + \lambda}$ is the mathematical score of the Right child node if the split is executed. $\frac{(G_L + G_R)^2}{H_L + H_R + \lambda}$ is the score of the original Parent node before any split occurs. It simply aggregates the total gradients and Hessians of all elements combined. λ is the L_2 regularization hyper-parameter sitting in the denominators: by adding a positive weight, XGBoost shrinks the overall score of a node. This prevents splits that isolate tiny, highly volatile subsets of observations, serving as an algorithmic buffer against overfitting. γ is the complexity penalty or user-defined threshold for tree pruning. Since splitting a node adds a new leaf to the tree structure, it costs an explicit penalty of γ . Since the algorithm only considers whether a value is above or below the threshold t (a rank-based decision), the absolute scale of the feature X is irrelevant. A monotonic transformation such as

$\log(X)$ or $X \times 100$ does not change the data's ordering, resulting in identical split points. While an LMM requires explicit interaction terms (e.g., $X_1 \times X_2$), XGBoost captures these naturally by nesting splits. If a split on Institutional Risk Index is followed by a split on Network Prestige, the model has effectively captured a non-linear interaction between these two variables.

1.2.3 R Packages

The empirical analysis was entirely executed using the R version 4.4.1 (R Core Team, 2024). Data preprocessing was based on the tidyverse ecosystem (Wickham et al., 2019), and readxl (Wickham & Bryan, 2023) for importing spreadsheet data. Global economic indicators were retrieved through the World Development Indicators database via the WDI package (Arel-Bundock, 2021). Spatial and geographical data processing incorporated rnatualearth and rnatualearthdata (Massicotte & South, 2023) to source map layers, while geographic distances and spatial geometries were calculated using the geosphere package (Hijmans, 2022). Furthermore, network-based properties of the data were explored through igraph (Csardi & Nepusz, 2006). Inequality and concentration metrics were computed using the ineq library (Zeileis, 2014). High-dimensional linear regressions with multi-way fixed effects were estimated using the computationally efficient fixest framework (Berge, 2021). LMMs relied on lme4 (Bates et al., 2015), with statistical significance evaluated through lmerTest (Kuznetsova et al., 2017). Conditional distribution dynamics were assessed through quantile regressions implemented in quantreg (Koenker, 2025). Bivariate data interpolation for irregularly spaced data points was handled via akima (Akima & Gebhardt, 2022). Finally, the tabular outputs of the statistical models were tidied and converted into standard data frames using broom (Robinson et al., 2024). Predictive modeling was implemented using the Extreme Gradient Boosting algorithm via xgboost (Chen et al., 2024), where the initial data partitioning and cross-validation workflows were managed through rsample (Silge et al., 2024). Data visualizations were generated using Qlik and ggplot2 (Wickham, 2016) in conjunction with scales (Wickham & Seidel, 2023) for axis transformations. Multiple graphical panels were assembled cleanly into unified layouts using patchwork (Pedersen, 2024), while interactive, web-based visual plots were rendered via plotly (Sievert, 2020).

CHAPTER II

DESCRIPTIVE STATISTICS AND DATA VISUALIZATION

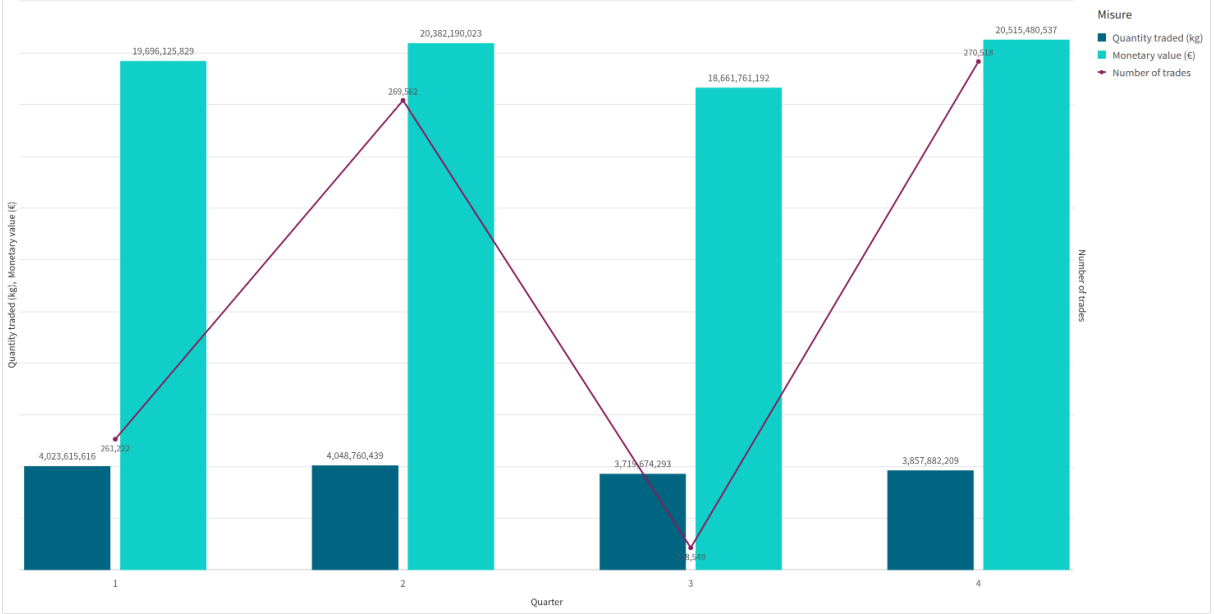
2.1 Micro and Macro-Dynamics: Firm-Level and Country-Level Data Profiles

Before formally estimating the structural determinants of Veneto's export performance via the Linear Mixed Model (LMM), a rigorous exploratory data analysis was conducted to map the underlying topology of the bilateral trade network. This section provides a multidimensional descriptive analysis of the 2024 granular cross-sectional data, disaggregating macroeconomic trade flows into their temporal, firm-level, geographical, and regional components. By systematically visualizing physical throughput (quantity traded), economic magnitude (monetary value), and network connectivity (transactional frequency and out-degree), we uncover extreme distributional asymmetries inherent in modern international trade. Specifically, this exploratory phase empirically identifies the profound market concentration driven by "superstar exporters," visually corroborates the gravity model of trade alongside the Alchian-Allen unit-value effect, and reveals significant intra-year cyclicity within the Veneto's manufacturing sector (*Ateco 2007, Section C, 10-33*). Ultimately, these preliminary diagnostics serve a dual purpose: they illuminate the structural archetypes and spatial heterogeneities of the Veneto's export base, while providing the fundamental empirical justification for the hierarchical econometric specification, variable transformations, and dimensionality reduction techniques employed in the subsequent regression analysis.

Figure 1 presents a longitudinal, dual-axis analysis of aggregate export dynamics across the four quarters of 2024. The visualization captures three fundamental dimensions of the macroeconomic trade network: physical throughput (Quantity traded in kg), economic magnitude (Monetary value in €), and network activity (Number of distinct trades). The data demonstrates a pronounced, synchronous deceleration across all three metrics during Q3. Physical volume contracts to an annualized nadir of 3.72 billion kg, accompanied by a proportional drop in monetary value (€18.66 billion) and transactional frequency (258,549 trades). This structural dip is highly characteristic of European and Italian manufacturing cycles, which typically experience severe output and export compressions during the summer months (e.g., August industrial closures). The most

analytically significant trend emerges in the fourth quarter. While the transactional frequency rebounds to a peak of 270,518 trades and the aggregate monetary value reaches its annualized zenith of €20.52 billion, the physical quantity traded (3.86 billion kg) fails to recover to its first-half baseline (which averaged over 4 billion kg). This divergence mathematically necessitates a substantive increase in the aggregate unit value (price per kilogram) of exported goods in Q4. This could indicate a compositional shift toward higher-margin, premium manufactured goods at the end of the fiscal year or delayed inflationary pass-throughs in export pricing. The trajectory of the network activity (number of trades, plotted on the secondary axis) closely mirrors the monetary value rather than physical volume. This suggests that the expansion of export value is driven more by an increased frequency of discrete commercial interactions (network densification) than by moving larger bulk shipments.

Figure 1. Historical Trends of Veneto Exports by Quarter of 2024



Note. Own elaboration based on ISTAT data (2024).

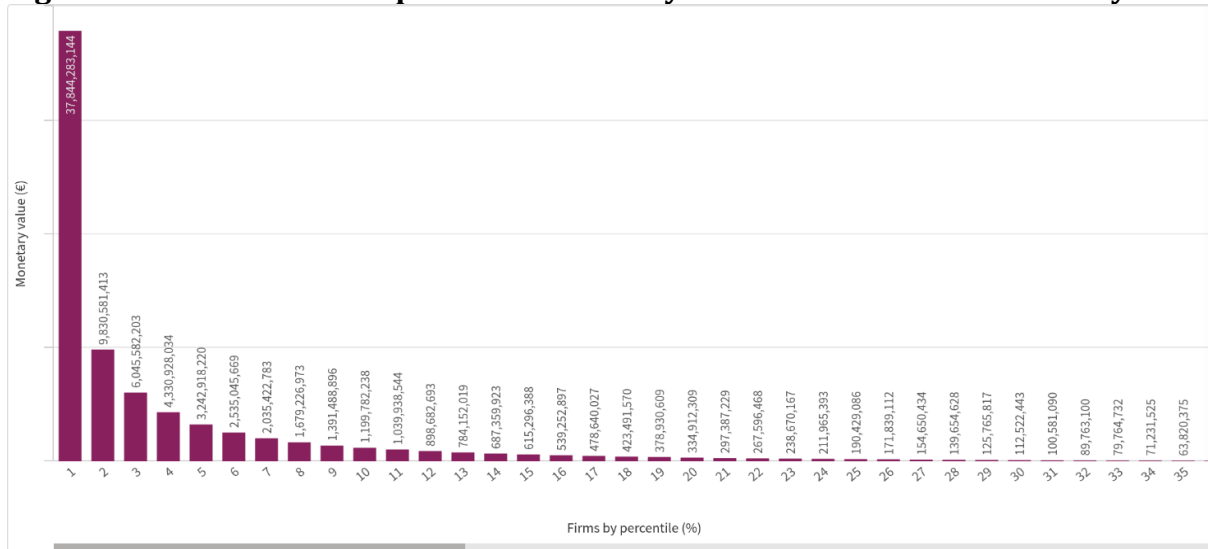
An exploratory analysis of the firm-level export data reveals extreme heterogeneity across the Veneto’s manufacturing base. By dissecting the aggregate monetary value, physical quantity, and geographical reach (*out-degree*) of the Veneto top-performing exporters, several distinct structural archetypes emerge within the trade network. To comply with stringent data privacy regulations (GDPR or Istat microdata protocols) and prevent the re-identification of specific high-performing economic agents, a cohort-based aggregation methodology was employed. Rather than plotting individual observations of

specific Veneto's firms, the latter were ranked and grouped into percentiles using a nested aggregation function:

$$P_i = 100 \times \frac{\text{Rank}_i}{N} \quad (5)$$

where i represents the specific firm. By presenting the total monetary value per percentile rather than the mean or individual values, the identity of any single "superstar exporter" (e.g., the leading firm in the 1st percentile) is mathematically obscured within the aggregate wealth of its elite peer group. This approach preserves the global variance and the shape of the distribution while ensuring that the "thin anonymity" of extreme outliers—implemented using IDs to refer to the firms, rather than their names—is replaced by robust, group-level confidentiality.

Figure 2. Barchart With Top Veneto's Firms by Percentile Based on Monetary Value

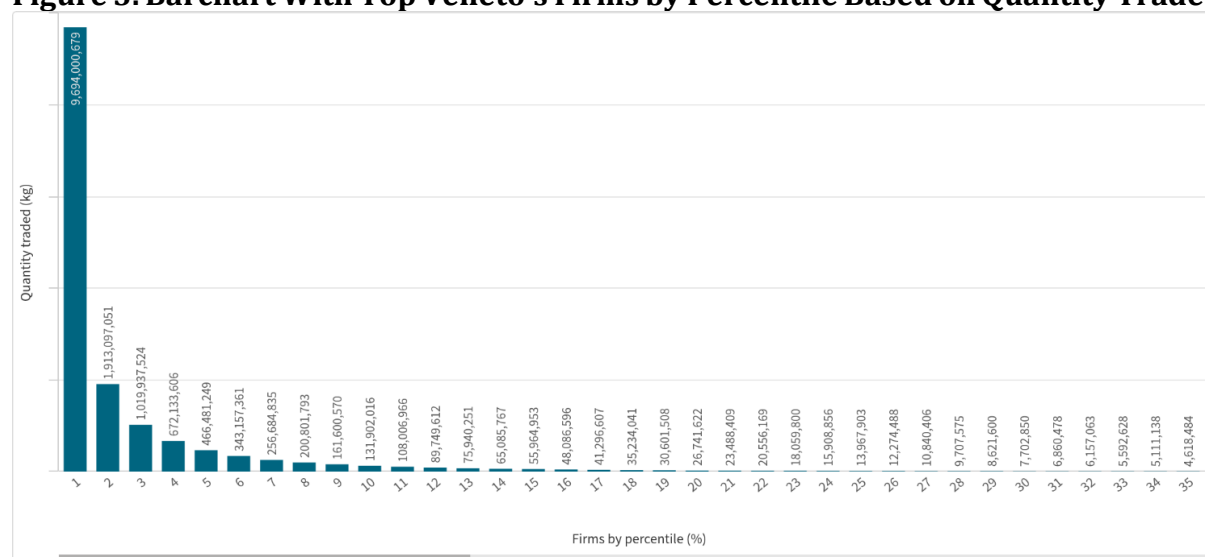


Note. Own elaboration based on ISTAT data (2024).

In Figure 2, the data reveals that the 1st percentile (the top 1% of Veneto's exporters) serves as the primary driver of regional trade, generating a total export monetary value of approximately €37.84 billion. This single centile's contribution is nearly four times greater than that of the 2nd percentile (€9.83 billion). This suggests that the head of the distribution is not merely a segment of the market but a separate economic regime entirely, likely comprised of multi-national industrial leaders with high R&D intensity and global network prestige. The transition from the 1st to the 5th percentile illustrates an exponential decay in market power. By the 5th percentile (€3.24 billion), the monetary

value has contracted by over 91% compared to the leaders. This steep gradient suggests that the barriers to reaching "superstar exporter" status are not linear; they involve significant sunk costs and productivity thresholds that only a fractional minority of the productive fabric can transcend. As the distribution moves toward the 35th percentile and beyond, the curve enters a stable long tail. The value at the 35th percentile (€63.8 million) represents an appropriate level of internationalization common to Veneto's robust SME sector, while the baseline level is represented by the 100th percentile occupied by firms that generate a total export monetary value of €135,759. While Veneto's firms are numerous, their aggregate economic weight is eclipsed by the top 1%, highlighting a dual economy structure within the region. The results underscore the [Melitz \(2003\)](#) paradigm of firm-level heterogeneity: export success is a "winner-take-most" game. From a policy perspective, the concentration observed in the 1st percentile implies that the regional economy's resilience is heavily dependent on the performance of a few hundred "hegemonic" firms. Conversely, the long tail suggests a vast field of firms that are active in international markets but operate with significantly lower scale and, likely, lower pricing power. A comparative analysis between the physical quantity distribution and export unit value distribution provides a granular structural mapping of the Veneto manufacturing sector, revealing the divergence between logistical mass and qualitative pricing power. By aggregating the firm population into centiles, these distributions expose the radical "superstar exporter" asymmetries that define the regional export vanguard for the 2024 fiscal year.

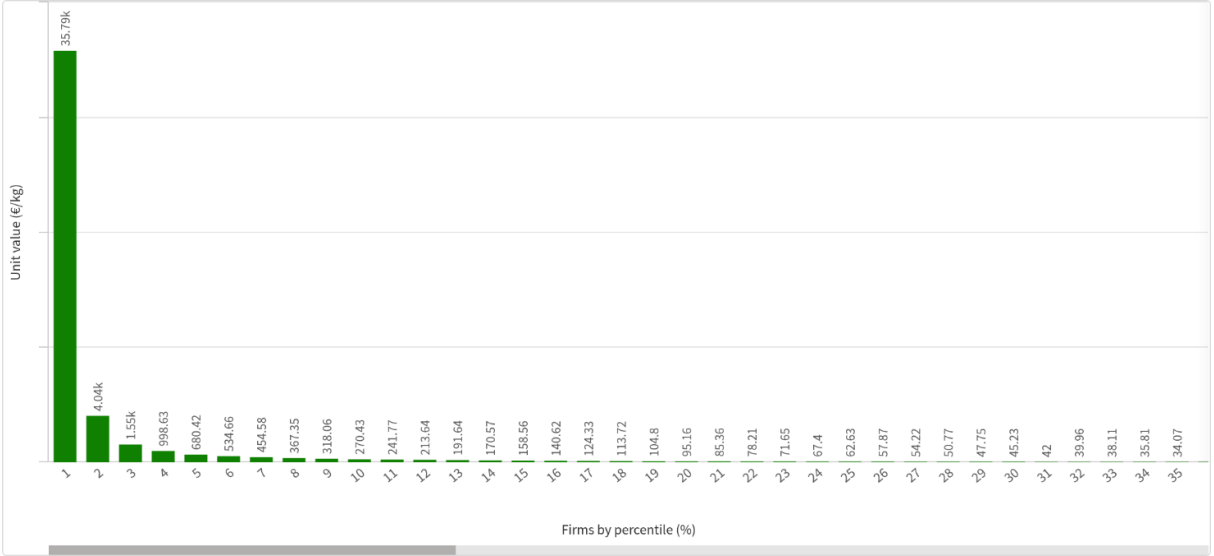
Figure 3. Barchart With Top Veneto's Firms by Percentile Based on Quantity Traded



Note. Own elaboration based on ISTAT data (2024).

In Figure 3, the distribution of quantity traded (kg) reflects the physical scale and industrial mass of the regional productive fabric: the 1st percentile handles a staggering 9.69 billion kg, representing the logistical backbone of the region. The transition to the 2nd percentile (€1.91 billion kg) marks a 5-fold reduction in volume. This indicates that the logistical infrastructure of Veneto’s trade is heavily dominated by a narrow cohort of industrial giants likely dealing in bulk commodities, raw materials, or heavy industry³. By the 10th percentile, the quantity has contracted to 131.9 million kg, less than 2% of the top centile’s volume. This confirms that for most Veneto’s firms, international trade is a low-volume activity, emphasizing the role of the 1st percentile as the primary driver of regional freight and logistical demand.

Figure 4. Barchart With Top Veneto's Firms by percentile based on unit value



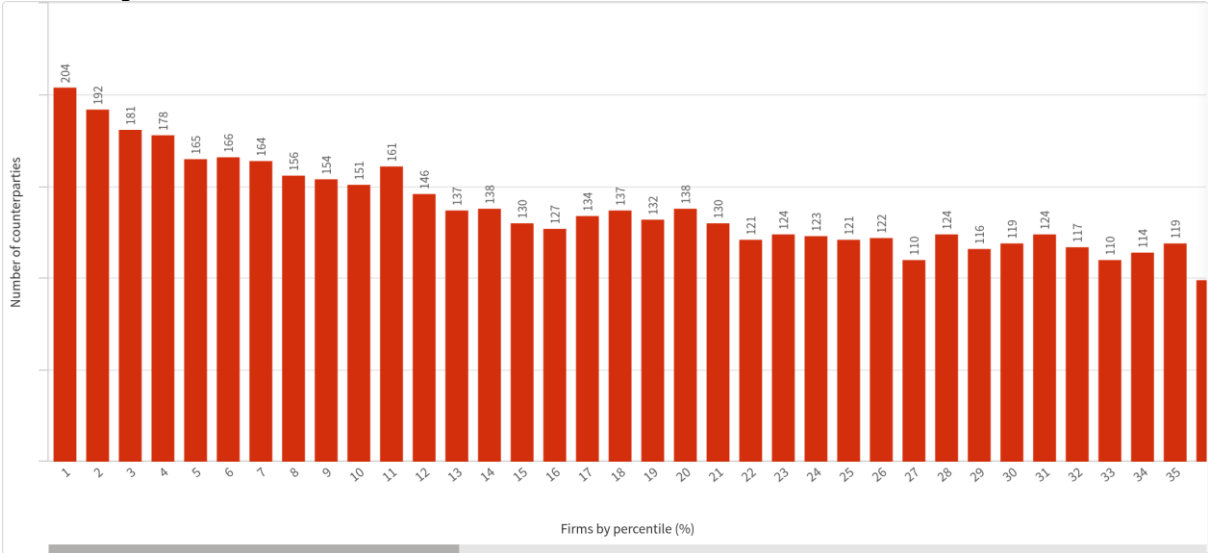
Note. Own elaboration based on ISTAT data (2024).

In Figure 4, the distribution of export unit value (€/kg) serves as a proxy for the technological frontier and vertical differentiation of Veneto’s exports. The 1st percentile achieves an extraordinary unit value of 35.79k €/kg. This represents a radical departure from the rest of the distribution; the 2nd percentile commands only 4.04k €/kg. This 9-fold drop-off is nearly twice as steep as the decay observed in quantity, suggesting that "superstar exporter" status in terms of pricing power is significantly more exclusive than "superstar exporter" status in terms of scale. Beyond the 10th percentile (€270.43/kg),

³ For the fiscal year 2024, Istat microdata identifies specific industrial sub-sectors with exceptionally high volume: *Iron and Steel Industry – Manufacture of Iron, Steel, and Ferroalloys* with 0.95 €/kg (Ateco code 24.10.0), *Paper and cardboard manufacturing* with 1.36 €/kg (Ateco code 17.12.0), *Soft drinks, mineral waters and other bottled waters industry* with 0.62 €/kg (Ateco code 11.07.0).

the unit value stabilizes into a long tail of standardized manufacturing. It demonstrates that the superstar phenomenon in the Veneto region is driven by quality sorting: these firms are exporting high-complexity goods (likely jewellery or precision machinery) that carry a significant "Made in Italy" price premium. This visualization serves as the empirical justification for the Alchian-Allen *"shipping the good apples out"* conjecture. These top-performing firms represent the absolute pinnacle of product quality. In a gravity model context, these are the only entities capable of fully absorbing high institutional risk and geodetic distance frictions, as their massive margins render transportation and transaction costs negligible as a percentage of total value.

Figure 5. Barchart With Top Veneto's Firms by Percentile Based on Number of Counterparties

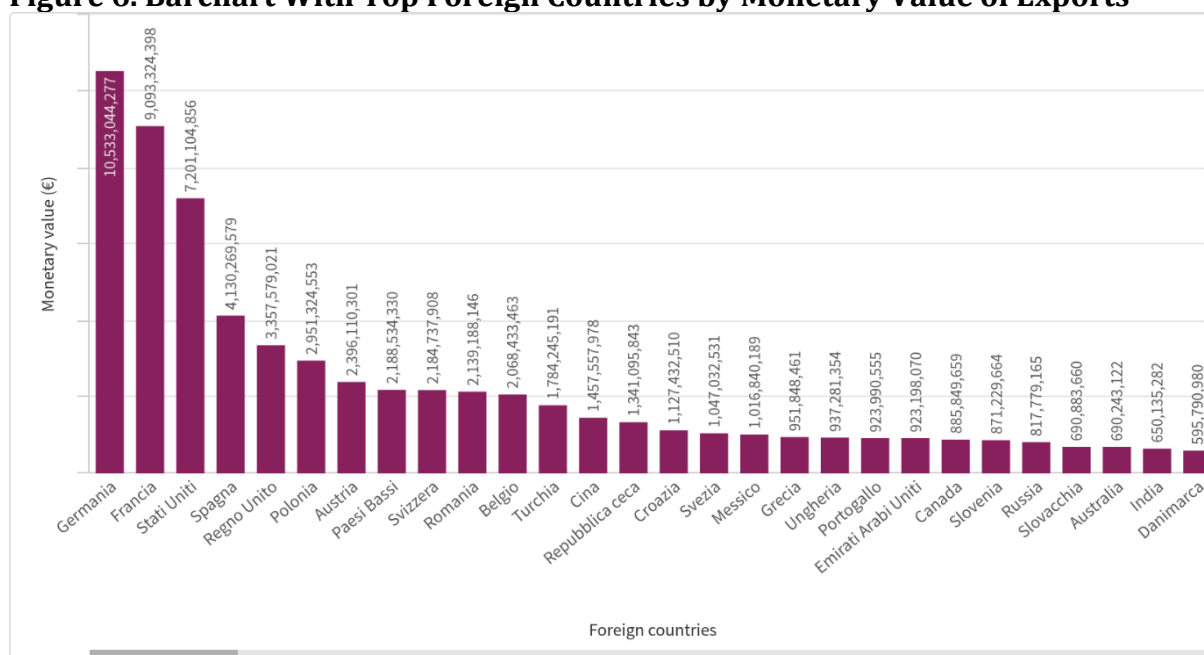


Note. Own elaboration based on ISTAT data (2024).

The analysis of Figure 5, representing the distribution of the number of counterparties (unique destination markets or trading partners) across Veneto’s firm percentiles, provides a critical topological counterpoint to the previous value-based distributions. In trade network theory, this metric represents the *out-degree* of the firm, serving as a direct measure of the extensive margin of trade. The most significant takeaway from this distribution is its relative linearity compared to the extreme exponential decay (power-law distribution) observed in monetary value, quantity, and unit value. The topological structure of Veneto’s export base is characterized by a 'fat-headed' distribution of counterparties; this high degree of connectivity across the first 35 percentiles implies a robust, decentralized network. Even if the 'superstar' leaders were removed, the region would retain a vast web of pre-existing commercial relationships. This suggests that

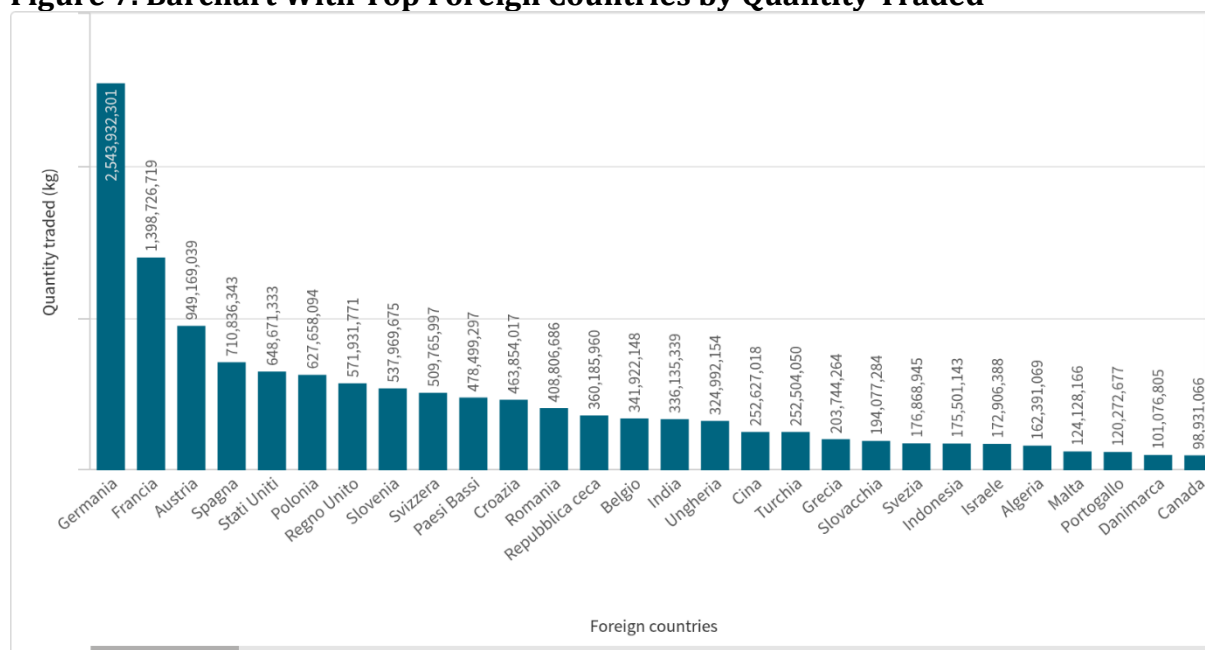
Veneto's competitive advantage lies in its relational infrastructure—the sheer number of active bridges between the region and the global marketplace—even if the high-value traffic across those bridges is concentrated in the hands of the ‘superstar exporters’. For policymaking, this implies that the challenge for Veneto is not getting firms to export to more countries (as they are already doing so) but rather increasing the qualitative value and pricing power of the trade flows within those established network connections. These findings robustly validate the econometric modelling strategy. First, the severe power-law distributions in both value and quantity necessitate the logarithmic transformations applied in the Linear Mixed Model (LMM) to normalize variance and prevent heteroscedasticity. Second, the disconnect between the firms with the most counterparties and the firms with the highest revenues justifies the inclusion of both *out_degree* (to measure raw market breadth) and *eigenvector centrality* (to measure the network prestige or quality of the destination markets) in the regression model, as sheer geographical spread does not guarantee maximum economic return. An examination of the destination markets for Veneto's exports reveals a highly concentrated, structurally asymmetric geographical footprint. By comparing the aggregate monetary value (Figure 6) against the physical quantity imported across foreign countries (Figure 7), critical insights regarding market integration, spatial proximity, and export composition emerge.

Figure 6. Barchart With Top Foreign Countries by Monetary Value of Exports



Note. Own elaboration based on ISTAT data (2024).

Figure 7. Barchart With Top Foreign Countries by Quantity Traded



Note. Own elaboration based on ISTAT data (2024).

Both distributions forcefully corroborate the standard Gravity Model of international trade, which suggests that bilateral trade flows are directly proportional to economic mass and inversely proportional to geographic distance. The Veneto's export ecosystem is overwhelmingly anchored by its core European Union partners. Germany stands as the undisputed primary nexus, absorbing over €10.5 billion in monetary value and a staggering 2.54 billion kg in physical volume. France strictly follows as the secondary hub (€9.09 billion; 1.39 billion kg). Together, these two immediate, high-GDP neighbour states account for a massively disproportionate share of the total network flow, underscoring the deep structural integration of Italian manufacturing within the continental European supply chain. A comparative cross-sectional analysis of the rank-order between the two plots reveals a critical decoupling of physical throughput and economic value, driven largely by geographic distance and transport costs. The distant premium market is the United States that ranks third in aggregate monetary value (generating over €7.20 billion) but drops to fifth in physical quantity (approximately 648 million kg). This severe divergence mathematically dictates a uniquely high unit value (value-to-weight ratio) for American exports. Economically, this aligns with the Alchian-Allen theorem (*"shipping the good apples out"*): the high per-unit transportation costs associated with transatlantic shipping disincentivize the export of bulk, low-margin goods, filtering the flow almost exclusively toward premium, high-value-added sectors (e.g., luxury goods, advanced

machinery, pharmaceuticals). Conversely, immediate geographic neighbours exhibit the inverse structural profile. Austria ranks third globally in physical volume absorbed (949 million kg) but drops significantly to seventh in monetary value (€2.39 billion). Similarly, Slovenia ranks eighth in volume (537 million kg) but falls well outside the top tier in monetary value. This spatial decay in unit value highlights these adjacent markets as primary destinations for bulk, heavy, or intermediate goods (e.g., base metals, construction materials, agricultural bulk) where minimal land-transport costs allow low-margin exports to remain economically viable. The pronounced heterogeneity across destination markets provides robust empirical justification for the inclusion of country-level random effects (1 | Paese) in the Linear Mixed Model. The baseline capacity of a country to absorb Veneto's exports is fundamentally dictated by unobserved structural characteristics; namely, geographic proximity, bilateral transport costs, and specific market demand for either premium or bulk goods. Treating the destination market solely as a uniform node without controlling for this severe volume-value asymmetry would introduce profound omitted variable bias into the estimation of export determinants.

While the standard Gravity Model remains the cornerstone of trade analysis, the contemporary global landscape necessitates an extension of this framework to account for Nearshoring and Friendshoring. These phenomena represent a strategic recalibration of Global Value Chains, shifting the focus from pure cost-efficiency to supply chain resilience and institutional security. Nearshoring reflects a strategic pivot toward regionalization, where firms relocate manufacturing and sourcing activities to geographically proximal nations. In the context of the Gravity Model, this goes beyond the mere minimization of geodetic distance; it is an attempt to mitigate spatial friction by ensuring operational agility. By shortening supply chains, exporters can reduce lead times and exposure to the systemic shocks inherent in transoceanic logistics, effectively leveraging geographic proximity as a buffer against global volatility. Simultaneously, the paradigm of Friendshoring (or "ally-shoring") introduces an ideological dimension to the gravity equation. This concept posits that trade flows are increasingly dictated by institutional proximity—the degree of political, legal, and credit alignment between trading partners. In this framework, geodetic distance is often secondary to geopolitical level. Nations with shared democratic norms and security alliances experience accelerated trade flows, as these shared frameworks act as non-tariff accelerants that

reduce the perceived risk of "supply chain weaponization" and expropriation. To rigorously integrate these concepts, the classical gravity equation can be augmented to include Geopolitical Affinity ($A_{\{ij\}}$) and Regional Resilience ($R_{\{ij\}}$):

$$T_{ij} = G \frac{M_i^\alpha M_j^\beta}{D_{ij}^\gamma} \cdot e^{\delta A_{ij} + \theta R_{ij}} \quad (6)$$

where T_{ij} represents the bilateral trade flow (monetary value) between origin i and destination j . G is a constant gravitational parameter. M_i and M_j are the economic masses (GDP) of the origin and destination markets. D_{ij} is the geodetic distance between the two centroids. A_{ij} stands for the Geopolitical Affinity index (Friendshoring) that represents institutional alignment. R_{ij} is the Regional Resilience factor (Nearshoring) that is a proxy for logistics integration. $\alpha, \beta, \gamma, \delta, \theta$ are parameters to be estimated, representing the elasticities and impact of each respective variable.

2.2 Spatial Distribution and Network Topology of Veneto's Export Base

Figure 8 illustrates the cross-sectional export performance across distinct Veneto provinces (Vicenza, Verona, Treviso, Padova, Venezia, Belluno, and Rovigo) mapped along physical throughput (quantity traded) and economic magnitude (monetary value). An analysis of this spatial distribution reveals a pronounced decoupling between trade volume and value extraction, indicative of divergent regional industrial specializations:

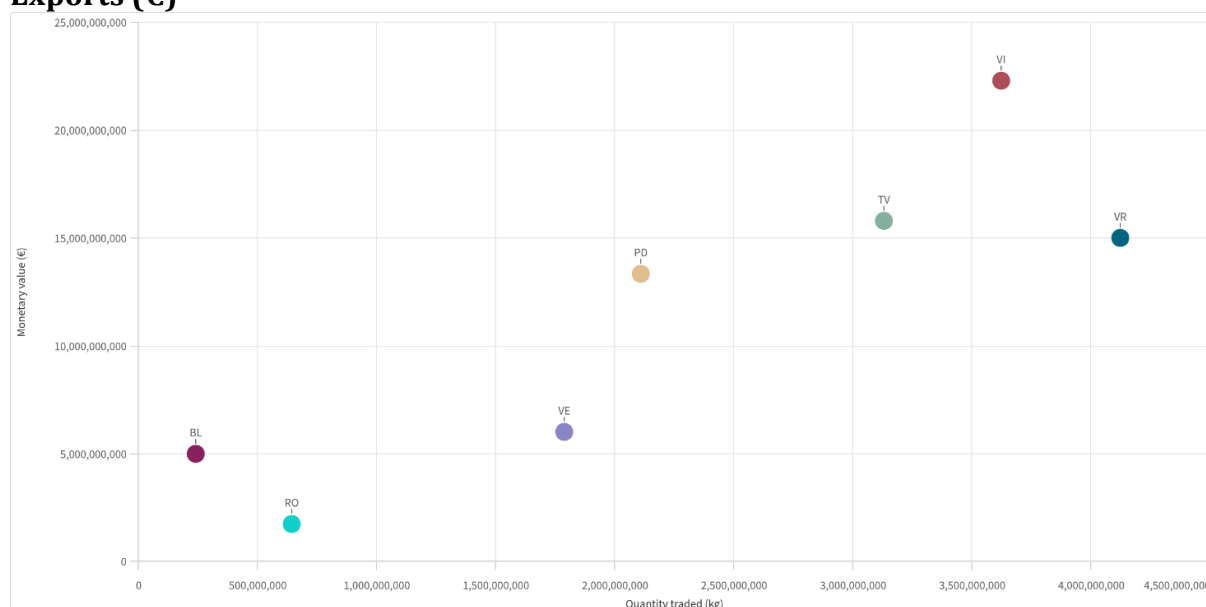
- Vicenza emerges as the undisputed leader in economic value generation, exceeding €22 billion in aggregate export value. However, it ranks second in physical quantity (approximately 3.6 billion kg). This positioning dictates an exceptionally high aggregate unit value (value-to-weight ratio). Economically, this suggests that the provincial industrial district is heavily concentrated in high-value-added, lightweight manufacturing sectors (e.g., goldsmith industry and production of advanced textiles).
- Conversely, Verona exhibits the inverse structural profile. It maximizes physical export throughput, leading the cohort with over 4.1 billion kg traded, but generates a substantially lower aggregate monetary yield (approximately €15 billion) relative to Vicenza. This trajectory implies a regional specialization in high-volume, lower-margin

bulk exports, potentially anchored in heavy industry, construction materials, or agribusiness (e.g., production of heating and air conditioning systems)

- Provinces such as Treviso (TV) and Padova (PD) form a mid-tier core, balancing robust volume with proportional monetary returns. Meanwhile, Belluno (BL) and Rovigo (RO) occupy the structural periphery of this specific regional network, characterized by highly constrained physical and monetary export footprints.

This spatial variance empirically justifies the use of hierarchical or mixed modelling frameworks. The capacity to generate export value is not uniformly distributed across the regional territory but is intrinsically tied to localized, sub-regional industrial ecosystems.

Figure 8. Bubble-Chart of Provinces by Quantity Traded (kg) and Monetary Value of Exports (€)



Note. Own elaboration based on ISTAT data (2024).

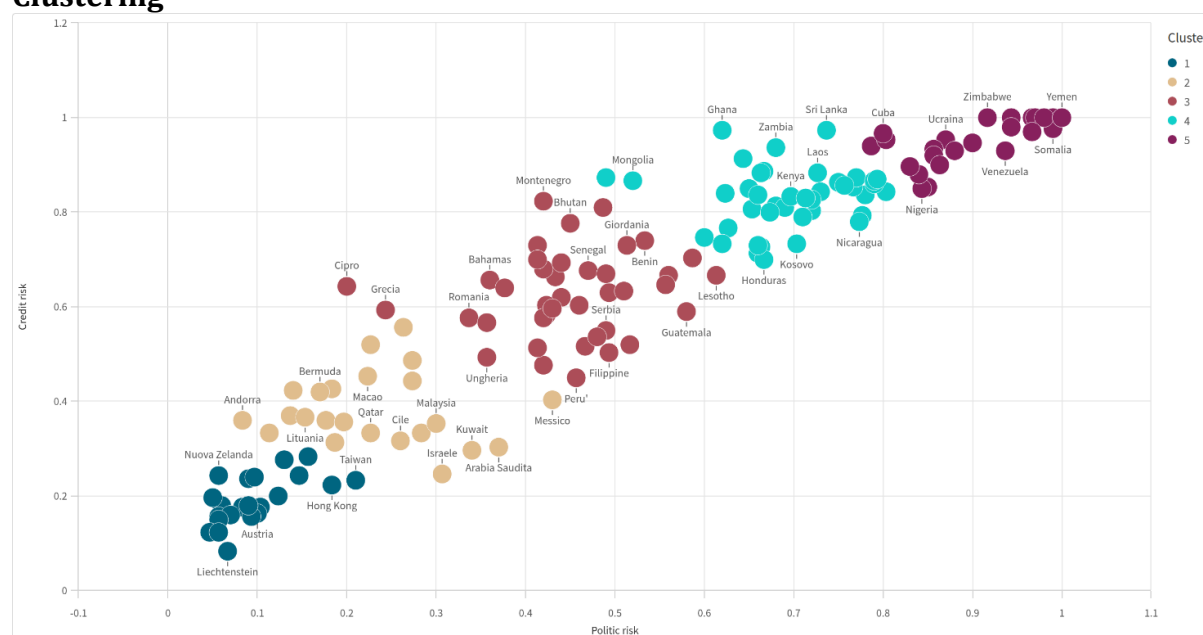
Figure 9 presents a diagnostic mapping of international destination markets based on their institutional fragility, plotting Political Risk⁴ against Credit Risk⁵. To identify structural market archetypes, an unsupervised machine learning algorithm (hierarchical clustering with k-means) was deployed, segmenting the global export network into five distinct risk strata. The scatter plot demonstrates a severe, near-linear positive covariance between political instability and credit default risk. This visually corroborates the earlier correlation heatmaps, confirming that political deterioration in a destination market almost strictly guarantees a corresponding escalation in financial friction. The

⁴ Average of Political violence, Transfer and Expropriation risks

⁵ Average of Sovereign, Banking and Corporate risks

lower-left quadrant (dark blue and beige clusters) identifies low-friction markets characterized by robust institutional frameworks (e.g., Austria, New Zealand, Liechtenstein). These destinations represent the "safe core" of the export network, minimizing transactional uncertainty for Veneto's firms. The central cluster encapsulates emerging economies (e.g., Philippines, Bahamas, Jordan) with moderate institutional volatility. These markets represent pivotal growth nodes but require exporters to navigate intermediate credit and political frictions. The upper-right quadrant (purple and light blue clusters) isolates markets suffering from extreme institutional fragility and systemic default probabilities (e.g., Ukraine, Yemen, Venezuela, Zimbabwe). Engaging with these nodes exposes exporters to maximum transactional friction, likely depressing aggregate export volumes or necessitating severe risk premiums on exported goods.

Figure 9. Foreign Countries Grouped by Political Risk and Credit Risk With K-Means Clustering

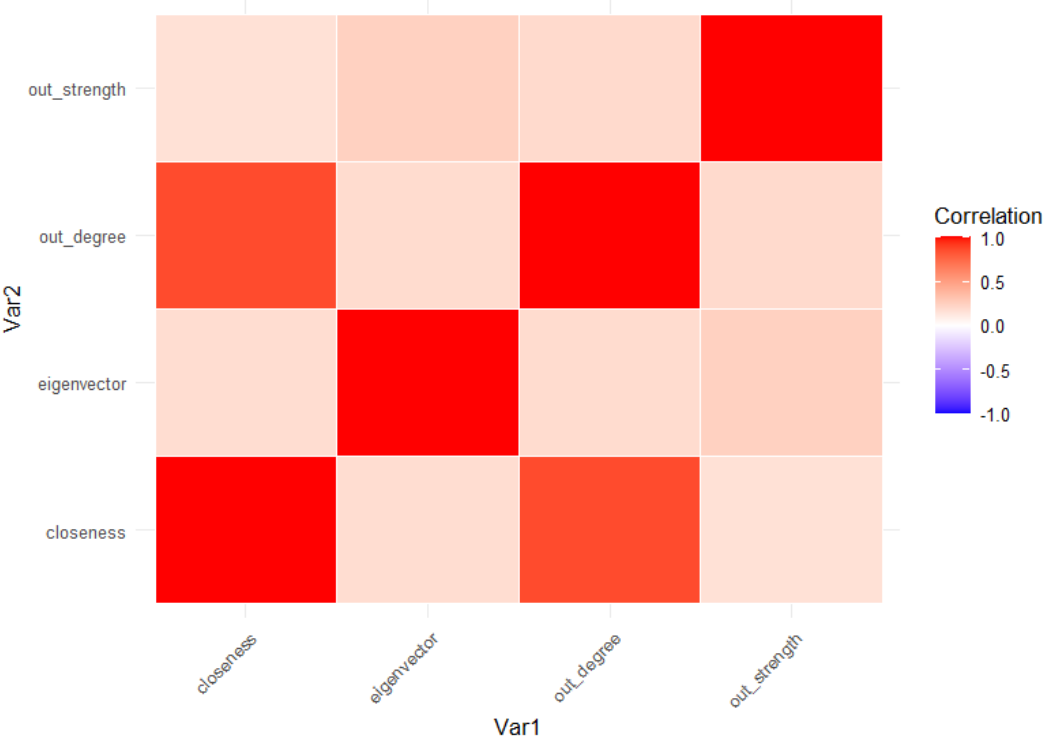


Note. Own elaboration based on SACE data (2024).

Political risk serves as a primary driver of financial instability: as a state's commitment to the rule of law degrades, the legal enforceability of debt contracts becomes increasingly precarious, raising the probability of corporate and banking defaults. Furthermore, political volatility frequently triggers capital flight and currency devaluation; transfer risk directly impairs the ability of both sovereign and corporate entities to service external debt obligations denominated in foreign currencies. Consequently, credit risk is rarely an idiosyncratic financial phenomenon; it is a lagging indicator of macro-institutional decay.

The clear stratification of the global market via k-means clustering provides profound theoretical support for the econometric models. It empirically validates the deployment of the PCA-derived "Country Risk Index", as the linear trajectory of the clusters proves that political and credit risks move in tandem. A 93.49% correlation confirms that these indicators are effectively capturing different facets of the same underlying latent variable of institutional quality. Navigating from Cluster 1 to Cluster 5 implies a compounding penalty on trade viability, which is precisely what the negative coefficient of the *Risk Index* in the LMM mathematically captures. To evaluate the structural properties of Veneto's exporting firms within the global trade network, an initial correlation analysis of four distinct centrality measures (*out_degree*, *closeness*, *eigenvector*, and *out_strength*) was conducted. The resulting correlation heatmap reveals critical insights into the topological dimensions of the network.

Figure 10. Correlation Heatmap of Centrality Measures Computed Through igraph Library

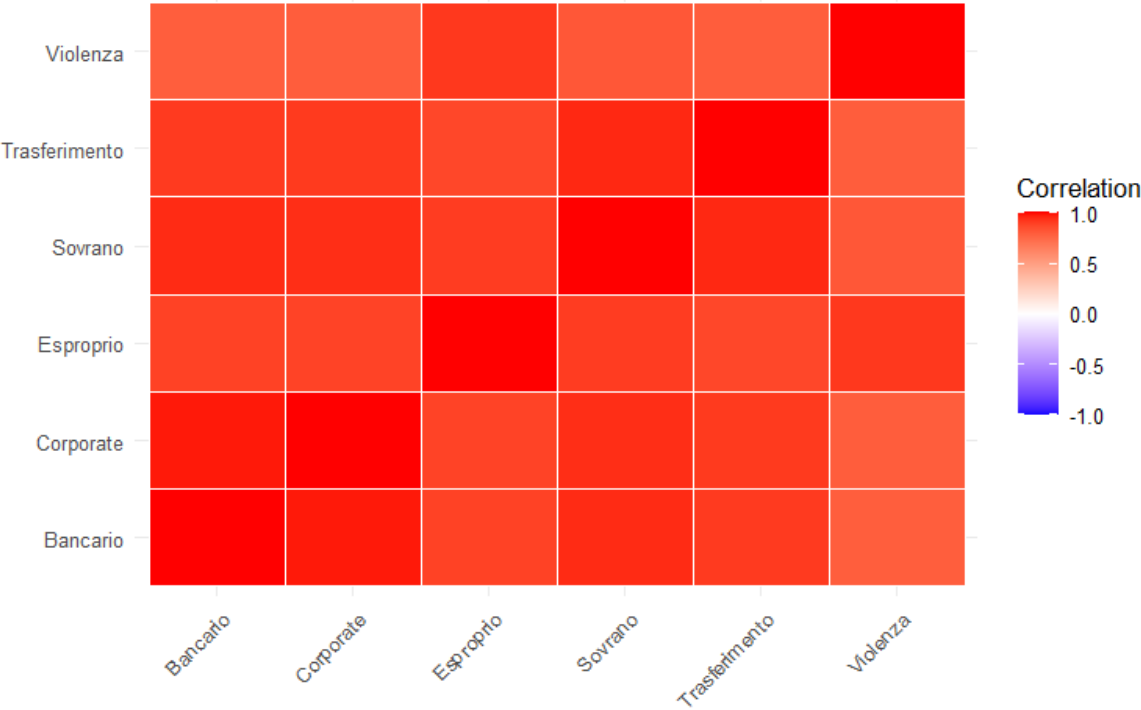


Note. Own elaboration based on ISTAT data (2024).

In Figure 10, the heatmap demonstrates a strong positive correlation (indicated by the dark orange/red shading) between *out_degree* and closeness centrality. Theoretically, this is consistent with bipartite trade network dynamics: firms that export to a high absolute number of distinct foreign markets (*out_degree*) naturally exhibit shorter

average path lengths to the rest of the global network (*closeness*). Consequently, including both metrics in a regression framework would introduce structural multicollinearity, as they largely capture the same latent construct of "market reach." Conversely, eigenvector centrality measure exhibits remarkably low correlation with both *out_degree* and *closeness* (represented by the pale/white shading). This mathematically confirms that a firm's sheer volume of connections does not dictate the *quality* of those connections. Eigenvector centrality successfully isolates a distinct topological dimension: "network prestige" or integration with core global hubs. These diagnostics justify the strategic selection of regressors for the econometric models. By retaining *out_degree* (to capture export breadth), *out_strength* and *eigenvector* (to capture export prestige) while discarding *closeness*, the model avoids Variance Inflation Factor (VIF) penalties while preserving multidimensional network effects.

Figure 11. Correlation Heatmap of Country Risk Indicators Collected From SACE



Note. Own elaboration based on SACE data (2024).

In Figure 11, the second heatmap visualizes the pairwise Pearson correlation coefficients across six macroeconomic and political risk indicators: Banking, Corporate, Expropriation, Sovereign, Transfer, and Political Violence. The correlation matrix is characterized by a nearly uniform, dense red gradient across all off-diagonal

intersections. This indicates pervasive and severe positive multicollinearity among the institutional risk variables. For instance, banking risk and corporate risk are virtually indistinguishable in their variance, and sovereign risk moves in strict tandem with transfer and violence risks. Injecting these raw, highly collinear variables into a Linear Mixed Model (LMM) would severely violate the Gauss-Markov assumptions regarding independent regressors, leading to inflated standard errors, unstable coefficients, and spurious p-values. This visual evidence robustly defends the methodological decision to deploy Principal Component Analysis (PCA). By compressing these variables into a single, orthogonal "Country Risk Index" (the first principal component which explains almost 91% of the latent variance of institutional friction), the model mathematically resolves the multicollinearity constraint while retaining the maximum informational variance of the destination country's risk profile.

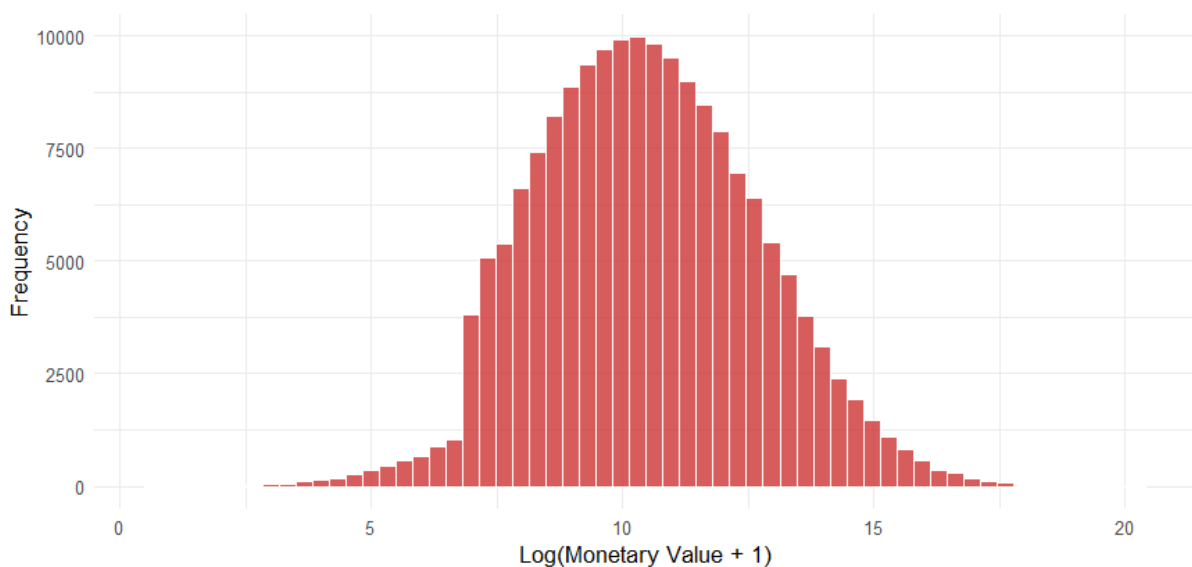
CHAPTER III

MODELLING THE DETERMINANTS OF TRADE VALUE AND QUALITY

3.1 Econometric Model With Export Value as Response Variable

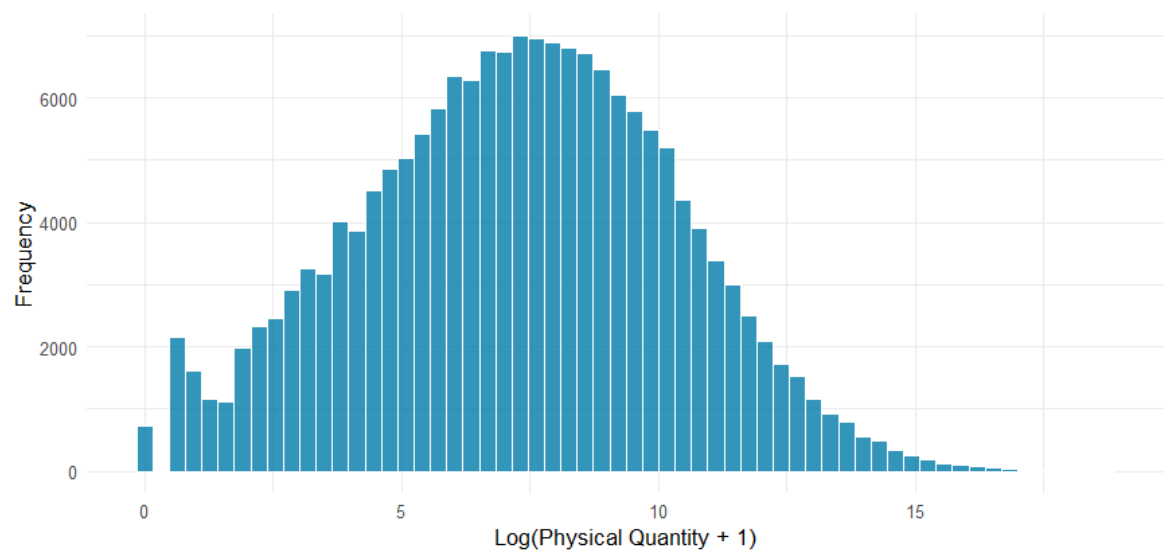
A preliminary inspection of the cross-sectional dataset further informs the data pre-processing strategy: the raw data for *total_val* (monetary value) and *total_qua* (quantity) exhibits extreme right-skewness, typical of gravity models of trade. For example, Firm 6's export value to Austria exceeds 2 million euros, whereas Firm 5's export value to Austria is 567 euros. This empirically supports the necessity of applying logarithmic transformations to the trade volume variables (Figure 12 and Figure 13) and foreign country related variables (GDP, GDP per capita and geographic distance) to reduce heteroscedasticity and allow for elasticity interpretations in the final model coefficients. Logarithms, in particular, are useful because they are more interpretable: changes in a log value are relative (percent) changes on the original scale. The presence of missing values (NaN) across all risk indicators for specific countries (e.g., Slovakia) necessitates a rigorous missing-data protocol. Listwise deletion (complete-case filtering) was applied prior to executing the PCA to ensure matrix computations execute properly without distorting the resultant eigenvector loadings.

Figure 12. Logarithmic Distribution of Monetary Value of Exports (€)



Note. Own elaboration based on ISTAT data (2024).

Figure 13. Logarithmic Distribution of Physical Quantity of Exports (kg)



Note. Own elaboration based on ISTAT data (2024).

Figure 14. Econometric Model With Export Value (€) as Response Variable

Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's method ['lmerModLmerTest']
Formula: $\log(\text{total_val}) \sim \log(\text{Distance_km}) + \log(\text{GDP_Eur}) + \text{Risk_Index_Standardized} + \text{out_degree_std} + \text{out_strength_std} + \text{eigen_std} + (1 \mid \text{firm_ID}) + (1 \mid \text{Paese})$
Data: final_model_data

AIC	BIC	logLik	deviance	df.resid
703847.5	703948.0	-351913.7	703827.5	171846

Scaled residuals:

Min	1Q	Median	3Q	Max
-5.8027	-0.6208	0.0319	0.6572	5.1248

Random effects:

Groups	Name	Variance	Std.Dev.
firm_ID	(Intercept)	0.94358	0.9714
Paese	(Intercept)	0.08992	0.2999
Residual		3.15491	1.7762

Number of obs: 171856, groups: firm_ID, 22818; Paese, 171

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	3.42517	0.42597	157.17476	8.041	2.02e-13 ***
log(Distance_km)	-0.27711	0.02704	141.08889	-10.247	< 2e-16 ***
log(GDP_Eur)	0.34255	0.01529	180.78840	22.404	< 2e-16 ***
Risk_Index_Standardized	0.12297	0.02979	162.53583	4.128	5.83e-05 ***
out_degree_std	0.69544	0.01468	9621.85027	47.372	< 2e-16 ***
out_strength_std	0.30709	0.01633	7627.49924	18.805	< 2e-16 ***
eigen_std	0.32995	0.02201	7068.92033	14.988	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

(Intr)	lg(D_)	lg(GDP_	Rs_I_S	ot_dg_	ot_st_	
lg(Dstnc_k)	-0.422					
log(GDP_Er)	-0.854	-0.107				
Rsk_Indx_St	-0.241	-0.278	0.433			
out_dgr_std	0.018	-0.006	0.007	0.007		
ot_strngth_	-0.001	0.002	0.000	-0.001	-0.097	
eigen_std	-0.002	-0.003	0.003	0.002	-0.225	-0.305

Note. Own elaboration based on ISTAT, SACE and WDI data (2024).

To estimate the structural determinants of Veneto's export value, a Linear Mixed Model was fitted using maximum likelihood estimation based on 171,856 bilateral trade flows, 22,818 firms and 171 foreign countries. In figure 14, the model specifies the natural logarithm of export value as the dependent variable, incorporating random intercepts for both the exporting firm (*firm_ID*) and the destination country (*Paese*) to control for unobserved baseline heterogeneity (e.g., some firms are just inherently better at exporting, and some destination countries are inherently harder to export to, regardless of the risk indicators). By including the random intercepts (1 | *firm_ID*) and (1 | *Paese*), the model is explicitly calculating and accounting for the intra-class correlation (ICC). For example, the model mathematically knows that rows belonging to Firm 4 are correlated with each other and adjusts the baseline accordingly, since the dataset contains the multiple bilateral trade flows of each firm with the value and the quantity exported towards each foreign country.

The analysis of the random effects provides the most compelling empirical evidence for the Melitz (2003) paradigm. The variance attributed to the firm-level ($\sigma^2\{\text{firm}\} = 0.94358$) vastly exceeds that of the destination country-level ($\sigma^2\{\text{country}\} = 0.08992$). This statistically confirms that success in international markets is primarily an idiosyncratic, firm-level phenomenon and it validates the "superstar exporter" paradigm: efficiency, innovation, brand equity, and product quality (the "Made in Italy" premium) are far more consistent in determining export value than the specific characteristics of the destination market. The high firm-level variance suggests that Veneto's export success is driven by a highly specialized vanguard of firms rather than a generalized regional trend. The fixed effects corroborate and extend the classical Gravity Model of trade:

- Economic Mass⁶: the positive coefficient ($\beta = 0.343$, $p < 2e-16$) indicates that export value is strongly and positively correlated with the destination market's GDP. This validates the fundamental gravity premise that trade flows are driven by the economic size of the trading partner.
- Spatial Quality Sorting⁷: the negative coefficient for distance ($\beta = -0.277$, $p < 2e-16$) provides empirical support for the Gravity Model theory. As geodetic distance between exporter and importer increases, we are 99,9% confident that the export

⁶ Logarithm of GDP expressed in Euro

⁷ Logarithm of geodetic distance expressed in Kilometres

value decreases, probably due to the cost of transportation that become more expensive. Consequently, Veneto's firms selectively "ship the good apples out," exporting higher-value-added products to distant markets to offset logistical frictions. This confirms that logistical costs and spatial barriers continue to penalize the total monetary volume of trade.

- The Standardized Risk Index exhibits a significant positive coefficient ($\beta = 0.123$, $p = 5.83e - 05$) which provides a robust empirical validation of the risk-return trade-off within the internationalization strategies of Veneto manufacturing sector. This positive estimate suggests that as riskiness of destination country increases by one standard deviation, there is a corresponding, measurable expansion in total export value. Far from indicating a preference for volatility, this confirms that institutional fragility—encompassing political uncertainty and credit instability—acts as a fundamental barrier that structurally alters the valuation of trade flows. From a Transaction Cost Economics (TCE) perspective, this phenomenon can be interpreted as the incorporation of risk premiums into the contractual valuation of goods. As highlighted by Rienstra-Munnicha and Turvey (2002), the increased probability of default or sovereign interference necessitates a strategic upward adjustment in monetary value to offset the increased costs of trade financing and insurance. Consequently, while Veneto's firms demonstrate remarkable systemic resilience, they are not immune to the frictions imposed by weak rule of law and financial volatility. The positive correlation further corroborates the Melitz-style selection effect in high-risk environments. Only the "superstar exporters"—those with the highest productivity and relational capital—possess the organizational capabilities to navigate the sunk costs and idiosyncratic hurdles of unstable markets. These elite firms effectively "charge for the risk," ensuring that the total monetary flow reflects not only the intrinsic value of the manufactured good but also the institutional premium required to sustain operations in emerging or volatile jurisdictions. This dynamic suggests a sophisticated market-selection mechanism where firms prioritize value extraction to compensate for the potential contraction of volume in institutionally distant territories.
- Market Reach (*out_degree_std*): displaying the highest magnitude among the standardized predictors ($\beta = 0.695$, $p < 2e - 16$) indicates that firms expanding

their extensive margin (the number of destination markets) capture significantly higher export value, suggesting that diversification is a primary driver of scale.

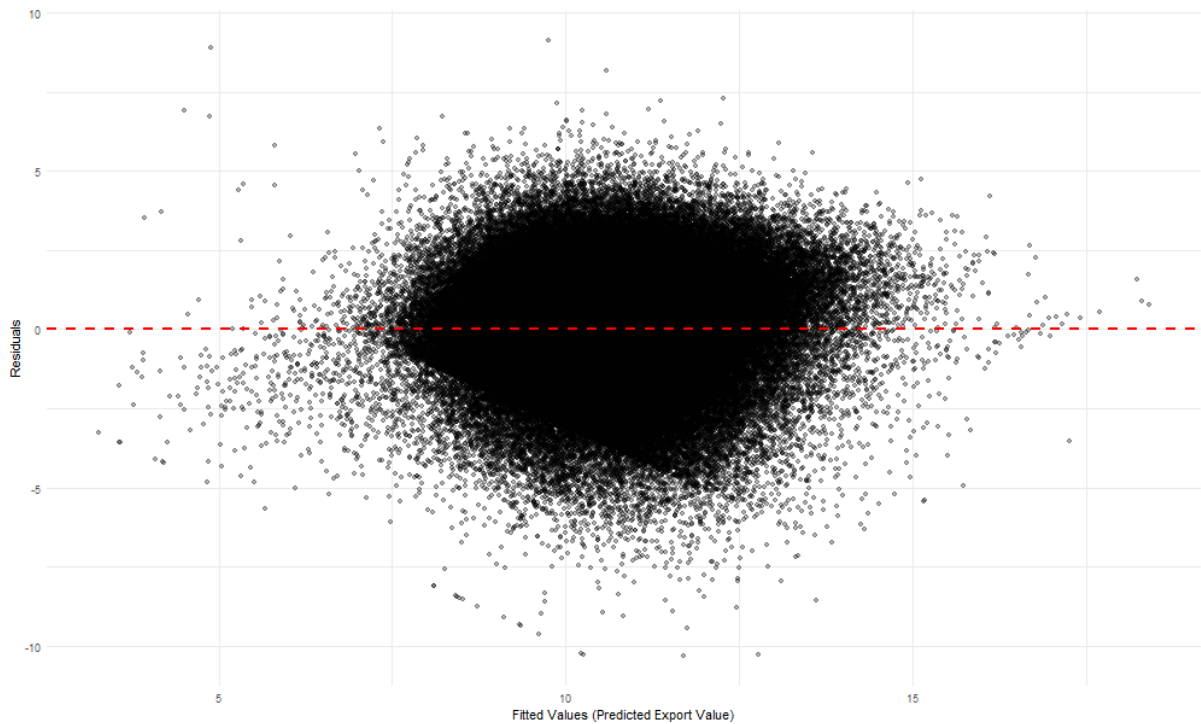
- Network Prestige (*eigen_std*): the significant positive coefficient ($\beta = 0.330$, $p < 2e - 16$) suggests a prestige premium. Firms integrated into high-prestige global hubs—exporting to well-connected countries—benefit from higher export values, likely due to their position in sophisticated Global Value Chains.
- Scale (*out_strength_std*): the positive and statistically significant coefficient for firm-strength ($\beta = 0.307$, $p < 2e - 16$) suggests that firms operating with high logistical intensity across their entire counterparty network possess a "scale advantage" that translates into higher monetary returns in specific bilateral flows. This reinforces the "superstar exporter" paradigm, demonstrating that the region's top performers are not only diversified in terms of destination markets (out-degree) but also operate at a scale of intensity that significantly boosts their total export value.

The model effectively demonstrates that Veneto's export performance is a multi-dimensional construct. It is anchored in firm-level excellence, modulated by geographic quality sorting, and increasingly dependent on topological positioning within the global trade network. The results suggest that for policymaking, fostering network integration and prestige may be as critical for value creation as traditional market-entry subsidies. The model exhibits high stability with an AIC of 703,847.5 and a clear convergence of residuals (median of 0.0319), indicating that the logarithmic transformations successfully neutralized the extreme right-skewness of the raw trade data.

To validate the Gauss-Markov assumptions underlying the LMM, standard diagnostic plots were generated from the model's residuals. In figure 15, the residuals are primarily clustered around the horizontal dashed red line at the zero-intercept. This centring suggests that the logarithmic transformation applied to the total export value is effective in capturing the non-linear relationship between the dependent variable and its predictors (geographic distance, GDP, Risk Index and network centrality metrics). While a minor diamond-shaped distribution of residuals suggests mild heteroscedasticity—common in high-frequency firm-level data—the concentration of errors within the ± 2 range for most observations indicates that the LMM successfully captured the systematic drivers of Veneto's export trade value. While heteroscedasticity can bias standard errors in simple OLS, the use of a LMM with random intercepts for firms and countries partially

mitigates this by partitioning the variance components. The extreme outliers observed, with residuals extending beyond ± 5 and even reaching ± 8 , corroborate the 'superstar exporters' hypothesis, identifying idiosyncratic firm-level successes that operate beyond the predictive frontier of standard macroeconomic variables. Points far below the line represent "under-performers": cases where the model predicted high value, but the actual flow was much lower, likely due to unobserved barriers such as specific trade sanctions, localized supply chain failures, or firm-specific distress. Finally, the extreme density of the cloud ($N = 171,856$) suggests that the model is asymptotically stable, meaning the estimates of the fixed effects (β coefficients) are highly reliable despite the presence of individual outliers.

Figure 15. Residuals vs Fitted Values Plot (Checking for Homoscedasticity)

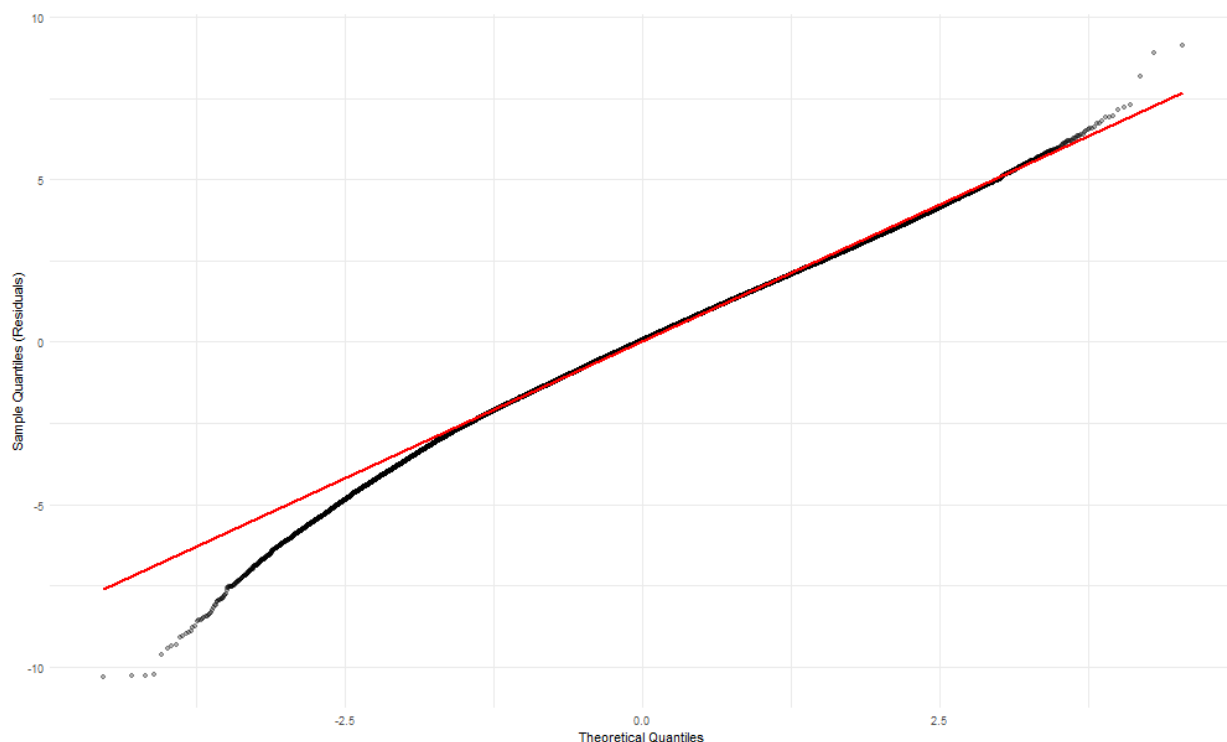


Note. Own elaboration based on ISTAT, SACE and WDI data (2024).

On the other hand, Figure 16 displays the Quantile-Quantile (Q-Q) plot that reveals a S-shaped distribution of residuals, characterized by heavy tails on both the positive and negative extremes. This deviation from the theoretical normal baseline is common in high-frequency bilateral trade microdata and reflects the presence of a fractional elite of firms whose performance is the result of exceptional, unobserved firm-level capabilities. However, given the large sample size, the Central Limit Theorem (CLT) ensures the asymptotic validity of the fixed-effect estimates. The model's consistency in the central

quantiles suggests a robust capture of the systematic drivers of trade value and errors are approximately normally distributed, while the tails identify the inherent volatility and heterogeneity of the Veneto manufacturing sector.

Figure 16. Normal Q-Q Plot (Checking for Normality of Residuals)



Note. Own elaboration based on ISTAT, SACE and WDI data (2024).

3.2 Econometric Model With Export Unit Value as Response Variable

To gain a deeper understanding of the competitive strategies employed by Veneto's manufacturing firms, the analysis shifts towards the estimation of unit values (€/kg). In contemporary trade literature, the unit value serves as a robust proxy for idiosyncratic quality and vertical differentiation. By shifting the dependent variable from total revenue to unit value of exports, the econometric model (Figure 17) effectively isolates the pricing power of the firm—defined as the capacity to command a price premium that is decoupled from transactional scale. Analysing this pricing power is statistically paramount: it allows for a clear distinction between "commodity-type" exporters competing on cost-efficiency and "frontier-type" exporters whose success is anchored in technical complexity, R&D intensity, brand equity and product quality. This methodological pivot is essential for identifying the true drivers of value creation within the Veneto's ecosystem, moving

beyond the observations of sheer market mass to capture the qualitative sophistication and resilience that define regional superstar status in the Global Value Chains.

Figure 17. Econometric Model With Export Unit Value (€/kg) as Response Variable

```
Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's method ['lmerModLmerTest']
Formula: log(unit_value) ~ log(total_qua) + log(Distance_km) + log(GDP_per_capita) +
  Risk_Index_Standardized + out_degree_std + out_strength_std + eigen_std + (1 | firm_ID) + (1 | Paese)
Data: final_model_data

            AIC          BIC      logLik   deviance  df.resid
401014.3    401124.9 -200496.2  400992.3    171845

Scaled residuals:
    Min       1Q   Median       3Q      Max
-13.2536  -0.4565   0.0181   0.4705  11.9767

Random effects:
Groups   Name              Variance Std.Dev.
firm_ID  (Intercept)  2.17760    1.4757
Paese    (Intercept)  0.01711    0.1308
Residual                    0.41399    0.6434
Number of obs: 171856, groups:  firm_ID, 22818; Paese, 171

Fixed effects:
              Estimate Std. Error      df    t value Pr(>|t|)
(Intercept)   3.971e+00  1.803e-01  1.677e+02  22.025  < 2e-16 ***
log(total_qua) -2.153e-01  7.863e-04  1.585e+05 -273.773  < 2e-16 ***
log(Distance_km)  3.138e-02  1.181e-02  1.288e+02   2.657  0.00889 **
log(GDP_per_capita)  4.773e-02  1.472e-02  1.753e+02   3.242  0.00142 **
Risk_Index_Standardized -3.617e-02  2.033e-02  1.443e+02  -1.779  0.07740 .
out_degree_std   3.346e-01  1.941e-02  1.880e+04  17.234  < 2e-16 ***
out_strength_std -3.685e-01  2.334e-02  1.814e+04 -15.784  < 2e-16 ***
eigen_std        2.513e-01  3.191e-02  1.808e+04   7.875  3.6e-15 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
              (Intr) lg(t_) lg(D_) l(GDP_ Rs_I_S ot_dg_ ot_st_
log(totl_q)   -0.039
lg(Dstnc_k)   -0.675  0.020
lg(GDP_pr_)   -0.839  0.000  0.186
Rsk_Indx_St   -0.609  0.006  0.007  0.824
out_dgr_std   0.092 -0.022 -0.002 -0.001  0.000
ot_strngth_   0.001 -0.030  0.000  0.000  0.000 -0.108
eigen_std     0.001 -0.004 -0.001  0.000  0.000 -0.226 -0.307
```

Note. Own elaboration based on ISTAT, SACE and WDI data (2024).

In Figure 17, the estimated coefficient (based on 171,856 bilateral trade observations, 22,818 firms and 171 foreign countries) provide a robust structural characterization of the regional export ecosystem, revealing how institutional frictions, geodetic distance and topological prestige modulate pricing power:

- consistent with the export value model, the variance decomposition of the random effects demonstrates that unobserved heterogeneity is overwhelmingly concentrated at the firm level ($\sigma^2\{\text{firm}\} = 2.17760$) rather than the destination country level ($\sigma^2\{\text{country}\} = 0.01711$). This mathematically confirms that export value and unit pricing power is an intrinsic characteristic of the firm. Unobservable factors such as brand equity, technological sophistication, and product quality (the "Made in Italy" premium) dictate the price per kilogram far more than the baseline characteristics of the foreign country that imported goods.

- **Institutional Risk:** the LMM model reveals a negative marginal effect ($\beta = -0.036$, $p = 0.07740$) on export unit value as riskiness of the foreign country—in terms of political and credit instability—increases by one standard deviation. While only significant at the 10% level, the directionality suggests that institutional fragility in destination markets acts as a qualitative friction, potentially forcing firms to downgrade their export mix towards risky nations. Indeed, poor institutional quality acts as a barrier that only the most resilient firms can bypass (Demir and Hu, 2020). This phenomenon of spatial product sorting reflects a defensive strategic posture adopted by most of the regional manufacturing base. To hedge against the potential loss of physical capital and financial investments in institutionally volatile jurisdictions, firms often opt to export standardized, lower-margin goods. This minimizes the Value at Risk (VaR) in the event of contractual defaults, expropriation, or sudden shifts in the regulatory environment. By shifting the export mix toward lower-complexity products, firms effectively reduce their exposure to the high sunk costs associated with premium-tier manufacturing in markets where the rule of law is inconsistently applied. However, this aggregate downward pressure on unit values masks a critical divergence driven by the "superstar exporters". While most of manufacturers may reduce the shipment of high-value-added goods towards risky nations, a fractional elite of hyper-productive firms possesses the organizational and financial depth to navigate the complex sunk costs of trade financing, credit insurance, and international legal hedging. For these firms, institutional risk does not necessitate a qualitative downgrade; rather, it justifies the application of significant risk premiums. Indeed, in this scenario the total export value tends to increase as we have seen in the previous econometric model, testifying that "superstar exporters" leverage their level of resilience and guide most of the value created in the GVCs.
- **Geographic Distance:** the positive coefficient ($\beta = 0.031$, $p = 0.00889$) provides empirical validation for the Alchian-Allen effect. As geodetic distance increases, the relative weight of fixed per-unit transportation costs favours the export of higher-quality goods, effectively "shipping the good apples out" to distant markets. This phenomenon, theoretically grounded in the work of Hummels and Skiba (2004), suggests that transportation costs act as a specific tax. Because these costs are added to the factory-gate price, the relative price of high-quality goods compared to low-quality goods decreases as distance grows, incentivizing firms to select higher-priced,

premium items for distant destinations to maintain competitive margins. Consequently, the statistically significant positive elasticity observed in the unit value model confirms that for Veneto's manufacturers, distance serves as a qualitative filter.

- **Population Wealth:** In contrast to aggregate economic mass, destination country unit wealth—proxied by GDP per capita—exhibits a significant positive correlation with export unit value ($\beta = 0.048$, $p = 0.00142$). It provides empirical weight to the wealth effect, yet its interpretation extends beyond simple market-scale dynamics into the realm of Engel's Law: as wealth increases, a household's basic food needs are met first (although the absolute spending on food rises, the share of total income it represents drops), leaving a larger percentage of disposable income for other things like technological items or luxury goods. This marginal effect seems to find resonance also in the Linder hypothesis which posits that the intensity of bilateral trade in manufactured goods is primarily dictated by preference similarity, which is a function of comparable per capita income levels (Linder, 1961). In the specific case of Veneto manufacturing sector—heavily oriented toward high-value-added, vertically differentiated products—the Linder effect suggests that trade flows are structurally directed toward markets with highly sophisticated overlapping demand structures. Wealthier nations do not merely exert a "gravitational pull" due to their absolute size, but specifically due to their demand for quality. This is consistent with the presence of non-homothetic preferences⁸, whereas income rises, the consumer's marginal utility shifts away from basic commodities toward idiosyncratic, high-unit-value goods. Consequently, the positive GDP per capita coefficient captures a sophistication premium. The idiosyncratic quality and brand equity of Veneto's manufacturing vanguard find a natural resonance in high-income economies that share similar consumption frontiers. This alignment confirms that for the Veneto's "superstar exporter", market wealth acts as a critical filter: high-quality exports are directed to markets where the preference for vertical differentiation matches the firm's high-cost, high-quality production structure, effectively validating the Linder conjecture in the contemporary regional context.

⁸ Non-homothetic preferences represent consumer choices where the Marginal Rate of Substitution (MRS) is not constant along rays from the origin: as income increases, the MRS changes even if the relative price ratio (P_x/P_y) remains constant. This means consumers change their relative consumption of goods as they get richer and MRS depends on both the ratio of goods and the total budget level.

- **Network Centrality:** firms that command a wider geographical reach extract significantly higher export unit value. A one standard deviation increase in *out-degree* is associated with a 33,5% increase in export unit value. Highly internationalized firms are typically producers of highly differentiated, premium goods capable of absorbing the fixed costs of entering numerous diverse markets. Integration with core global hubs yields a distinct pricing premium. A one standard deviation increase in *eigenvector centrality* drives a 25,1% increase in export unit value. Firms trading with highly central, wealthy nodes (e.g., Germany, USA, France) cater to sophisticated consumer bases willing to pay a premium for Italian quality. Conversely, a firm's total global physical throughput (*out_strength_std*) exhibits a highly significant negative coefficient ($\beta = -0.369$, $p < 2e - 16$). Firms that move the heaviest aggregate volumes globally often sacrifice unit value for logistical scale, further cementing the structural divide between high-volume industrial/commodity exporters and low-volume luxury/specialized manufacturers.
- **Physical Quantity Interaction ($\log(\text{total_qua})$):** the model identifies a highly significant negative elasticity ($\beta = -0.215$, $p < 2e - 16$). This suggests a fundamental trade-off in the Veneto's export base: higher transactional volumes are structurally associated with lower unit values, indicating that mass exports typically reside in the more standardized, commodity-intensive segments of the value chain.

The econometric model confirms that Veneto's export pricing power is not a random distribution but a structured outcome of topological positioning and institutional navigation. The results suggest that for the region's manufacturing base, the highest value premiums are found at the intersection of high network prestige and low-volume, high-complexity specializations. The significance of geographic distance suggests that Veneto's "superstar exporters" have not overcome spatial frictions, and high-quality exports are primarily absorbed by affluent markets with similar preference structures.

Notwithstanding the explanatory power of the current specification, a critical caveat remains regarding the level of sectoral aggregation. The empirical analysis was conducted on the broad manufacturing landscape of the Veneto region, encompassing the entirety of *ATECO 2007, Section C (divisions 10–33)*. While this allows for a comprehensive regional overview, it introduces a significant degree of unobserved intra-sectoral heterogeneity that potentially masks more nuanced economic dynamics. By treating the manufacturing

sector as a monolithic entity, the econometric model assumes a shared baseline for unit values that may not exist in reality. The structural “price floor” of a product is dictated by its technical complexity and material scarcity. Without controlling for specific ATECO codes, the model’s error term is forced to absorb massive variances in product-specific price baselines, which may lead to a compositional bias in the estimated coefficients.

In the inferential phase, competing LMM models (Figure 18 and Figure 19) were compared via ANOVA, confirming that the extended Gravity Models offered a vastly superior fit over baseline models. Specifically, the Likelihood Ratio Tests performed on the nested models demonstrate that the inclusion of institutional risk and topological network metrics significantly enhances the explanatory power of both the total value and unit value specifications. By rejecting the null hypothesis that the additional predictors have zero coefficients, the ANOVA results provide the formal statistical foundation for the subsequent interpretation of "superstar exporter" dynamics and the regional resilience.

Figure 18. Models’ Comparison (Export Value as Response Variable)

```
Data: final_model_data
Models:
model_1: log(total_val) ~ log(Distance_km) + log(GDP_Eur) + (1 | firm_ID) + (1 | Paese)
model_2: log(total_val) ~ log(Distance_km) + log(GDP_Eur) + Risk_Index_Standardized + out_degree_std +
out_strength_std + eigen_std + (1 | firm_ID) + (1 | Paese)
      npar    AIC      BIC   logLik deviance Chisq Df Pr(>Chisq)
model_1     6 707417 707478 -353703   707405
model_2    10 703847 703948 -351914   703827   3578  4 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note. Own elaboration based on ISTAT, SACE and WDI data (2024).

Figure 19. Models’ Comparison (Export Unit Value as Dependent Variable)

```
Data: final_model_data
Models:
model_unit_1: log(unit_value) ~ log(total_qua) + log(Distance_km) + log(GDP_per_capita) +
(1 | firm_ID) + (1 | Paese)
model_unit_2: log(unit_value) ~ log(total_qua) + log(Distance_km) + log(GDP_per_capita) +
Risk_Index_Standardized + out_degree_std + out_strength_std + eigen_std + (1 | firm_ID) +
(1 | Paese)
      npar    AIC      BIC   logLik deviance  Chisq Df Pr(>Chisq)
model_unit_1     7 401555 401626 -200771   401541
model_unit_2    11 401014 401125 -200496   400992  548.83  4 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note. Own elaboration based on ISTAT, SACE and WDI data (2024).

CHAPTER IV

FORECASTING PRICING POWER THROUGH MACHINE LEARNING

4.1 Predictive Performance of the Unregularized XGBoost Algorithm

To bridge the gap between structural inference and predictive accuracy, the research framework incorporates a machine learning approach utilizing the Extreme Gradient Boosting algorithm. While the preceding LLMs effectively isolate the average elasticities of trade determinants, XGBoost is deployed to capture the non-linear interactions and complex decision boundaries inherent in high-dimensional trade microdata. This transition allows for an assessment of whether the idiosyncratic pricing power (export unit value) of Veneto's manufacturing firms can be forecasted with precision using a non-parametric, ensemble-based learning method. To ensure the robust generalization of the model and mitigate the risk of overfitting, a rigorous data splitting protocol was implemented. The observations were partitioned into three distinct subsets:

- In-Sample Training Set (60%): utilized for the primary learning phase, where the algorithm iteratively constructs decision trees to minimize the objective loss function.
- Out-of-Sample Validation Set (20%): employed for real-time performance monitoring and hyperparameter calibration, ensuring that the model does not become overly attuned to the noise within the training data.
- Hold-Out Test Set (20%): reserved exclusively for final evaluation. This subset provides an unbiased estimate of the model's predictive performance on novel, unseen data, representing the ultimate test of the algorithm's external validity.

Table 1. Predictive Performance of the Unregularized XGBoost Algorithm

Table: XGBoost Predictive Performance (Raw Scale: Euros/kg)

Dataset	MAE	RMSE	R2
In-Sample (Train 60%)	236.867	4692.578	0.749
Out-of-Sample (Validation 20%)	323.634	5247.145	0.651
Hold-Out (Test 20%)	324.565	5173.314	0.400

Note. Own elaboration based on ISTAT, SACE and WDI data (2024).

The results presented in the performance table (Table 1) reveal a nuanced narrative regarding the predictability of export unit values. The model demonstrates a significant

R^2 decay, falling from 0.749 in the training set to 0.400 in the hold-out test set. While an R^2 of 0.400 in out-of-sample trade microdata is respectable, the gap suggests a degree of overfitting inherent in an unregularized XGBoost architecture. The model effectively "memorizes" specific firm-level price points in-sample but finds it challenging to replicate that accuracy when confronted with the extreme heterogeneity of the broader manufacturing population. A critical diagnostic finding is the massive discrepancy between the Mean Absolute Error (MAE) and the Root Mean Square Error (RMSE): on average, the model's predictions deviate by roughly €325 (Test MAE) from the actual unit value; instead, the RMSE squares the error terms, thus it is hyper-sensitive to large deviations. The fact that the RMSE is nearly 16 times larger than the MAE is empirical evidence of the "superstar exporter" phenomenon. The algorithm can predict "normal" manufacturing unit values with reasonable accuracy, but it fails to capture the extreme "tails" of the distribution, those ultra-high-value jewellery or precision-tooling shipments that reach thousands of euros per kilogram. These outliers exert a disproportionate influence on the RMSE, highlighting the idiosyncratic nature of high-end "Made in Italy" pricing that defies standard algorithmic prediction. The performance results suggest that while macroeconomic variables and network metrics provide a solid baseline for prediction, the residual volatility in unit values is driven by firm-specific factors that are not fully captured even by sophisticated non-linear models. In an academic context, this reinforces the argument that Veneto's pricing power is not merely a function of "big data" predictors, but of unique, unobservable intangible assets—such as brand heritage and artisanal skill—that remain difficult to quantify algorithmically.

4.2 Predictive Performance of the Regularized XGBoost Algorithm

To refine the predictive architecture and address the systemic overfitting observed in the initial unregularized iteration, the research framework transitioned toward a regularized XGBoost configuration. While the previous model achieved high in-sample performance by "memorizing" idiosyncratic firm-level noise, this secondary approach prioritizes generalization—the ability of the algorithm to maintain consistent predictive power when confronted with novel data. In an academic context, this represents a deliberate shift from minimizing training error to maximizing structural robustness across the full distribution of Veneto's manufacturing exports.

The regularization strategy employs a multi-layered approach to enforce model parsimony, utilizing three specific mechanisms to prevent the algorithm from "over-learning" from the extreme outlier's characteristic of the "superstar exporter" paradigm:

- Complexity Control⁹: by reducing the learning rate and restricting tree depth, the model is forced to learn gradually from broad patterns rather than isolating specific high-value firms in deep terminal leaves. This ensures that the estimated rules reflect regional economic trends rather than firm-specific pricing anomalies.
- Stochasticity and Subsampling¹⁰: introducing randomness into the row and feature selection prevents the model from over-relying on a specific subset of variables (e.g., being overly dominated by the "Country risk Index" or "Prestige"). This stochastic gradient boosting forces the trees to discover diverse, redundant signals, enhancing the model's stability.
- Explicit Penalization¹¹: The implementation of L1 (Lasso) and L2 (Ridge) penalties on leaf weights, combined with a high minimum child weight equal to 20, acts as a mathematical barrier. The algorithm is prohibited from creating a predictive rule unless it applies to a significant cluster of observations, effectively filtering out the "noise" generated by idiosyncratic luxury or high-complexity shipments.

Table 2. Predictive Performance of the Regularized XGBoost Algorithm

Table: Regularized XGBoost Predictive Performance (Raw Scale: Euros/kg)

Dataset	MAE	RMSE	R2
In-Sample (Train 60%)	345.142	6237.957	0.557
Out-of-Sample (Validation 20%)	389.832	7245.431	0.335
Hold-Out (Test 20%)	370.488	5648.059	0.285

Note. Own elaboration based on ISTAT, SACE and WDI data (2024).

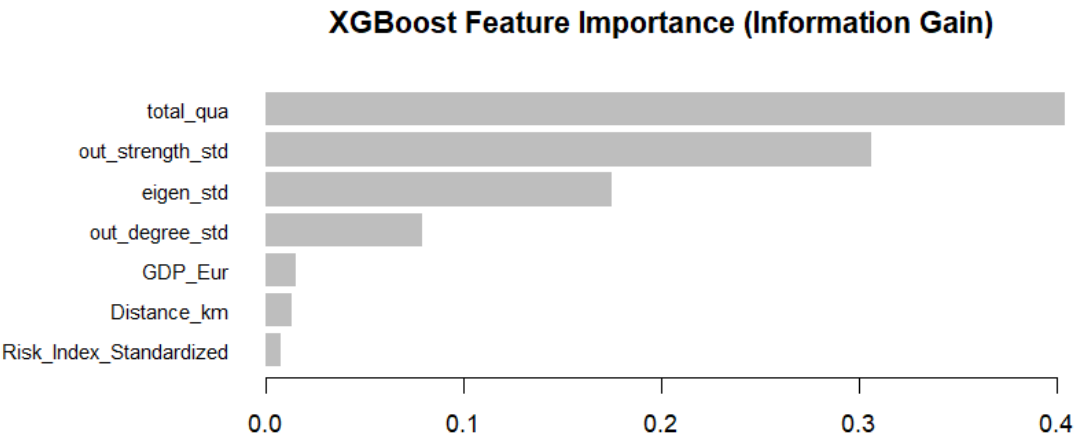
Table 2 illustrate the classic Bias-Variance Trade-off: while the absolute predictive power appears lower than the unregularized model, the structural integrity is significantly improved. The training R^2 dropped from 0.749 (unregularized) to 0.557, and the test R^2 settled at 0.285. Though lower in absolute terms, the gap between train and test performance has narrowed. This convergence indicates that the model is no longer chasing "hallucinated" patterns in the noise. The R^2 of 0.285 suggests that nearly 30% of

⁹ $\eta = 0.05$ and $max_depth = 3$
¹⁰ $subsample = 0.7$ and $colsample_bytree = 0.7$
¹¹ α (α), λ (λ) and $min_child_weight = 20$

the variance in raw unit values—an incredibly noisy variable—is predictable solely through the macroeconomic and network features identified in the LMM. The Test RMSE (5648.06) remains significantly higher than the Test MAE (370.49). In this regularized context, this discrepancy is not a sign of model failure, but a mathematical confirmation of the S-Shaped (Heavy-Tailed) distribution of the residuals discussed in the Q-Q plot analysis. Even with aggressive regularization, the quadratic nature of RMSE heavily penalizes the model for failing to predict the "superstar" jewellery or precision-tooling shipments. However, the stability of the MAE suggests that for the common Veneto manufacturer, the model’s error is manageable and consistent across all data splits.

The transition to a regularized framework confirms a fundamental truth about the Veneto export ecosystem: structural variables (GDP, Distance, Network Prestige) provide a solid baseline for mid-market goods, but "superstar" pricing power remains largely idiosyncratic. By forcing the model to be "shallower" and more "group-oriented", we have successfully isolated the systematic drivers of price from the unobservable firm-level characteristics. The regularized XGBoost serves as a high-fidelity baseline model; the remaining error represents the unobserved intangible capital (design, heritage, quality) that characterizes the "Made in Italy" brand and inherently resists algorithmic standardization.

Figure 20. XGBoost Feature Importance Plot



Note. Own elaboration based on ISTAT, SACE and WDI data (2024).

In Figure 20, the XGBoost Information Gain analysis corroborates the structural centrality of firm-level characteristics over destination-market macroeconomics. The

overwhelming importance of *total_qua* and *out_strength* confirms that the primary determinant of pricing power is the structural transition from mass-production to specialized-niche exporting. Furthermore, the superiority of *eigenvector centrality* over *out-degree* underscores that the prestige of a firm's network position is the most informative relational predictor of export unit value, while traditional gravity variables (GDP and Distance) function as secondary environmental moderators rather than primary drivers of idiosyncratic value. It is important to note that the low gain for the Country Risk Index does not contradict the LMM's finding that risk penalizes export unit value. Rather, it means that if we want to predict the exact price of a shipment, knowing the volume and firm's prestige is much more helpful than knowing the destination's risk level.

4.3 K-fold Cross Validation of the Error Metrics

To solidify the statistical reliability of the predictive framework and ensure that the estimated performance metrics are not an artifact of a specific data partition, the analysis employs k-fold Cross-Validation (CV). In an academic context, cross-validation serves as a rigorous stress test for the model's stochastic stability. By iteratively partitioning the dataset into k non-overlapping subsets—where each "fold" serves as the test set while the remaining k-1 folds facilitate training—the procedure minimizes the risk of selection bias and provides a more generalized estimation of the algorithm's out-of-sample performance.

Table 3. K-fold Cross Validation of the Error Metrics

Fold	MAE_Test	RMSE_Test	R2_Test
1	450.049	12835.320	0.229
2	342.476	4171.380	0.358
3	355.260	4425.248	0.367
4	397.661	8297.607	0.159
5	373.663	4887.859	0.288
AVERAGE (CV)	383.822	6923.483	0.280

Note. Own elaboration based on ISTAT, SACE and WDI data (2024).

The results of the 5-fold cross-validation, summarized in Table 3, offer a transparent view of the model's performance consistency across the heterogeneous landscape of Veneto's manufacturing sector. The average R^2 of 0.280 across the five folds aligns almost perfectly

with the previously observed hold-out test result of 0.285. This convergence is a critical indicator of model robustness: it proves that the regularized XGBoost architecture is capturing stable, systematic signals rather than idiosyncratic noise. However, the intra-fold variance in R^2 (ranging from 0.159 in Fold 4 to 0.367 in Fold 3) highlights the underlying "market turbulence" inherent in trade data. These fluctuations suggest that the predictive power of macroeconomic and network variables is partially dependent on the specific mix of sub-sectors and firm sizes present within each fold. The discrepancy between the Mean Absolute Error (MAE) and the Root Mean Square Error (RMSE) persists across all folds, serving as a persistent empirical signature of the "superstar exporter" phenomenon. Fold 1 presents a significant outlier in terms of error magnitude, with an RMSE of 12,835.32—nearly double the cross-validation average. Given that the MAE for Fold 1 (450.05) remains relatively stable, this spike in RMSE indicates the presence of a few extreme high-value observations within that specific subset that the model failed to predict accurately. In quadratic metrics like RMSE, a single "jewellery-sector" outlier (e.g., €5,000/kg) that is predicted at a standard manufacturing rate (e.g., €200/kg) creates a massive error penalty. The stability of the average MAE (383.82) suggests that for the vast majority of Veneto's manufacturing base, the model's predictions are highly consistent. The deployment of k-fold validation confirms that the model is neither "underfitting" nor "overfitting" in a way that compromises its utility. Instead, it defines the predictive frontier for Veneto's regional trade:

- **Generalization:** the consistency of the average metrics suggests that the model's findings—regarding the importance of network prestige, GDP, and institutional risk—are applicable across the entire regional population, not just a subset of firms.
- **Structural Volatility:** the high RMSE in specific folds (like Fold 1 and Fold 4) reinforces the earlier theoretical claim that high-end pricing power is an idiosyncratic firm-level asset. While the model can predict the "average" pricing of the manufacturing sector with nearly 30% accuracy, the "extreme upper tail" of the Made in Italy brand remains algorithmically elusive.

The cross-validation process validates the regularized XGBoost as a statistically stable instrument for trade forecasting. The model exhibits a reliable predictive floor, confirming that while global structural forces (gravity and topology) provide a significant predictive baseline, the ultimate source of value in the Veneto region resides in firm-level qualitative advantages that transcend standardized econometric estimation.

CHAPTER V

STRUCTURAL ANALYSIS OF THE VENETO'S STRATEGIC SECTORS

5.1 Methodological Framework

While the preceding chapters established the overarching macroeconomic and geographic framework of the Veneto region's export trade flows, this chapter shifts the analytical lens toward a granular, sector-specific dissection of the regional economy. The persistence of the "Made in Italy" brand in international markets is no longer sustained uniformly by traditional clusters. Instead, it is increasingly driven by a highly specialized, technologically advanced vanguard. The chapter isolates and evaluates the structural dynamics of the region's core industrial drivers, focusing specifically on three critical manufacturing domains defined by the classification of economic activities (ATECO):

- ATECO 10 (*Food industry*);
- ATECO 15 (*Manufacture of leather and similar articles*);
- ATECO 26 (*Manufacture of computers, electronics and optical products; electromedical apparatus, measuring apparatus and clocks*);
- ATECO 28 (*Manufacture of machinery and equipment*);
- ATECO 30 (*Manufacture of other means of transport, including shipbuilding, locomotives, railway rolling stock, aerospace vehicles, and military combat vehicles*);
- ATECO 32.1 (*Manufacture of jewellery, costume jewellery, and related articles; processing of precious stones*).

The theoretical justification for centring the empirical analysis on these specific sectors is grounded in the structural transformation of the Italian industrial district. As articulated by Corò and Volpe (2003), the phenomenon of production fragmentation has structurally redrawn the geographic boundaries of local economies. The traditional, self-contained neo-Marshallian cluster has evolved out of necessity. Under the pressures of globalized production networks, the local division of labour has integrated with international value chains, prompting a strategic shift toward high-value-added phases. This evolutionary trajectory is further formalized by Corò and Micelli (2006), who illustrate that survival in the global economy requires industrial districts to nurture "pivot firms". These leading entities serve as structural system integrators. In sectors like precision mechanics and advanced electronics, these firms capitalize on localized, tacit engineering expertise while simultaneously managing highly complex global supply pipelines. Consequently,

analysing ATECO codes 10, 15, 26, 28, 30 and 32.1 provides an empirical assessment of the exact nodes where Veneto's regional specialized knowledge intersects with international market demand. To systematically evaluate the competitive resilience of these strategic sectors, this chapter synthesizes the foundational technological classification framework of Lall (2000) with the structural patterns of technical change pioneered by Pavitt (1984). Lall's taxonomy categorizes manufactured exports based on their underlying technological intensity, skill requirements, and entry barriers, distinguishing sharply between low, medium, and high-technology goods (Table 4).

Table 4. Framework for the Vertical Differentiation of Goods

Technological Tier (Lall, 2000)	Structural Category (Pavitt, 1984)	Mapping to Veneto's Economic Activities
Low-Technology Products	Supplier-dominated: driven by low R&D intensity, simple assembly operations, and high labour-cost sensitivity. Highly vulnerable to price-based competition.	Traditional district baseline such as ATECO 10 (food industry) and ATECO 15 (leather items and similar articles).
Medium-Technology Products	Scale-intensive and specialized suppliers: characterized by skill-intensive engineering, complex process technologies, and extensive sub-assembly networks. Requires deep path-dependent expertise.	ATECO 28 (precision machinery & equipment), segments of ATECO 30 (automotive & railway components) and ATECO 32.1 (jewellery, costume jewellery, and precious stones).
High-Technology Products	Science-based: dominated by rapid product obsolescence, fast-evolving scientific frontiers, and massive capital reinvestment in R&D and advanced testing infrastructure.	ATECO 26 (computers, electronic, and optical products) and segments of ATECO 30 (aerospace and defence).

Note. Own elaboration based on Lall (2000) and Pavitt (1984).

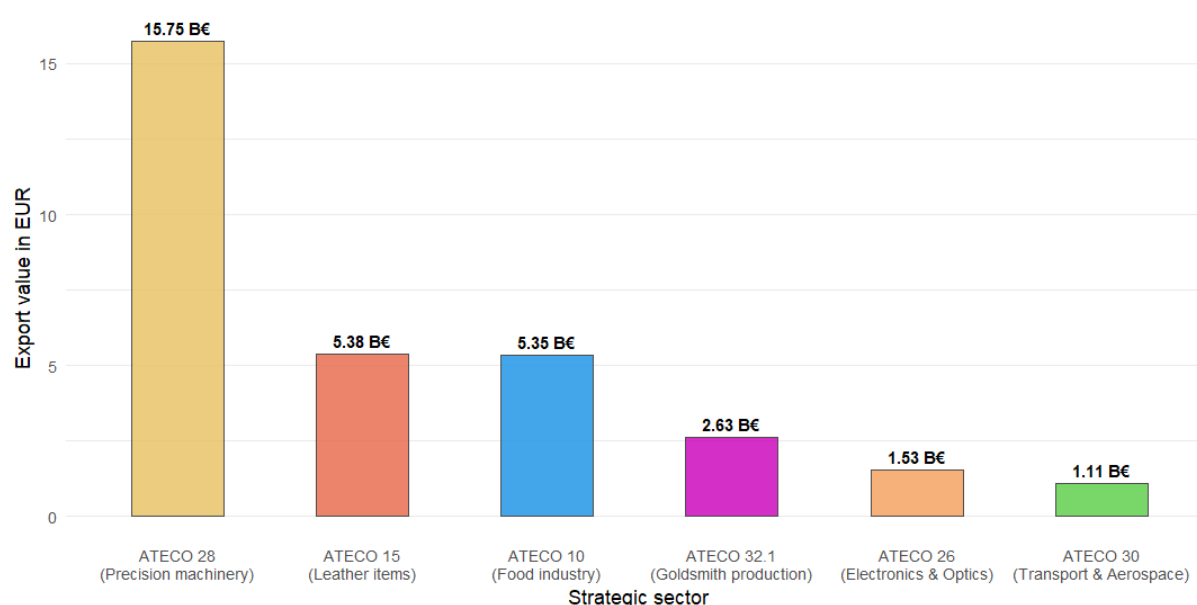
By separating the empirical models along these technological lines, this research investigates whether the determinants of export performance operate under distinct

structural regimes depending on a sector's technological complexity. Within this context, Sterlacchini (2001) notes that for Italian manufacturing, export performance is heavily contingent upon firm-level innovative intensity and R&D capabilities, rather than simple scale economies. Evaluating the structural resilience of heterogeneous firms within these advanced sectors requires a departure from traditional, mean-based linear estimation techniques. The empirical strategy implemented in this chapter relies on a dual methodological framework: Poisson Pseudo-Maximum Likelihood (PPML) and Quantile Regression (QR). The adoption of the PPML estimator is theoretically rooted in the modern gravity model literature, as synthesized by Shepherd et al. (2019). Traditional log-linearized Ordinary Least Squares (OLS) gravity estimations introduce severe distortions in the presence of heteroscedastic error terms—a condition ubiquitous in enterprise-level trade data. Because the variance of the error term typically scales with market size, log-linearization violates the exogeneity condition, yielding biased and inconsistent parameter estimates. Furthermore, traditional linear specifications fail to mathematically account for zero trade flows, dropping non-exporting firms and introducing selection bias. As highlighted by Shepherd et al. (2019), PPML addresses these limitations by estimating the gravity equation in its native multiplicative form, thereby preserving the structural validity of the error terms under heteroscedastic conditions; the model can incorporate zero-trade matrices, providing a statistically consistent assessment of the barriers governing market entry and exit. While PPML optimizes the estimation of aggregate sectoral trade gravity, it remains restricted to evaluating conditional mean expectations. To uncover the behavioural asymmetries hidden within the firm population, this chapter implements Quantile Regression. Given the deep right-skewness and Pareto-like distributions characteristic of export value aggregates, the "representative firm" paradigm is economically insufficient. Quantile Regression relaxes the assumption of a uniform parameter distribution by minimizing asymmetric absolute deviations. This allows for the estimation of distinct slopes (β_τ) across different percentiles of the export value distribution. Consequently, this methodology allows us to test whether institutional risk, geodetic distance, and network prestige exert symmetric pressures on small, vulnerable exporters (the lower quantiles) versus the hyper-productive "superstar exporters" (the upper quantiles) navigating the Global Value Chain. Finally, Chapter V deploys this unified structural approach to provide an empirical portrait of competitive resilience in the Veneto manufacturing sector.

5.2 Quantile Regression Models: Structural Non-Linearities in ATECO 28

By deconstructing the structural landscape of the Veneto's outbound trade mass—both in absolute financial terms (Billions of Euros) and relative macroeconomic distribution (percentage of total regional export value)—the plots expose a profound structural asymmetry. This polarization provides strong empirical validation for moving beyond homogeneous regional trade aggregates, shifting instead toward an analytical focus on the core strategic sectors that anchor the modern industrial district network.

Figure 21. Absolute Financial Weight of the Strategic Sectors in 2024

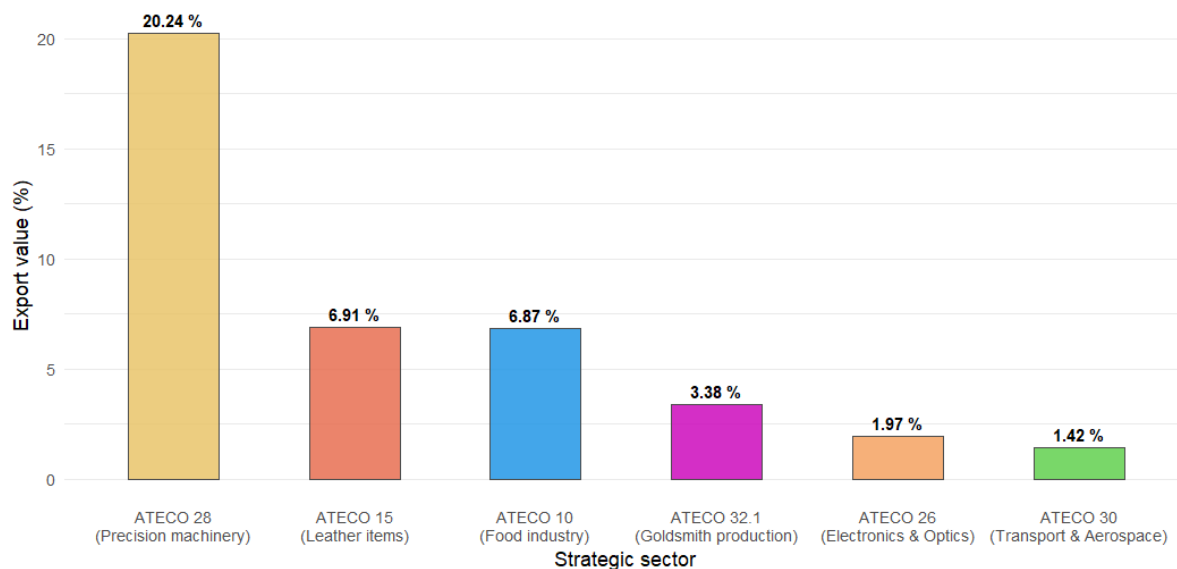


Note. Own elaboration based on ISTAT data (2024).

The descriptive hierarchy revealed by Figure 21 and Figure 22 illustrates that the regional manufacturing backbone is far from uniform: ATECO 28 (Precision Machinery) emerges as the dominant force within the regional export matrix. With an absolute trade volume of €15.75 Billion, it accounts for an astonishing 20.24% of the entire regional export base. This single sector constitutes over one-fifth of Veneto's outbound trade flows, demonstrating its role as a key driver of internationalization. ATECO 15 (Leather items) and ATECO 10 (Food industry) form a stable secondary tier, generating €5.38 Billion (6.91%) and €5.35 Billion (6.87%) respectively. These traditional sectors reflect the ongoing strength of the classic "Made in Italy" identity, anchoring local employment and traditional processing networks. Moving toward higher levels of technological complexity (Lall, 2000), ATECO 32.1 (Goldsmith production/Jewellery) commands €2.63 Billion

(3.38%), followed by ATECO 26 (Electronics & Optics) at €1.53 Billion (1.97%) and ATECO 30 (Transport & Aerospace) at €1.11 Billion (1.42%). While these high-tech clusters exhibit a smaller absolute financial footprint compared to precision machinery, they represent the region's qualitative and innovative frontier.

Figure 22. Percent Financial Weight of the Strategic Sectors in 2024

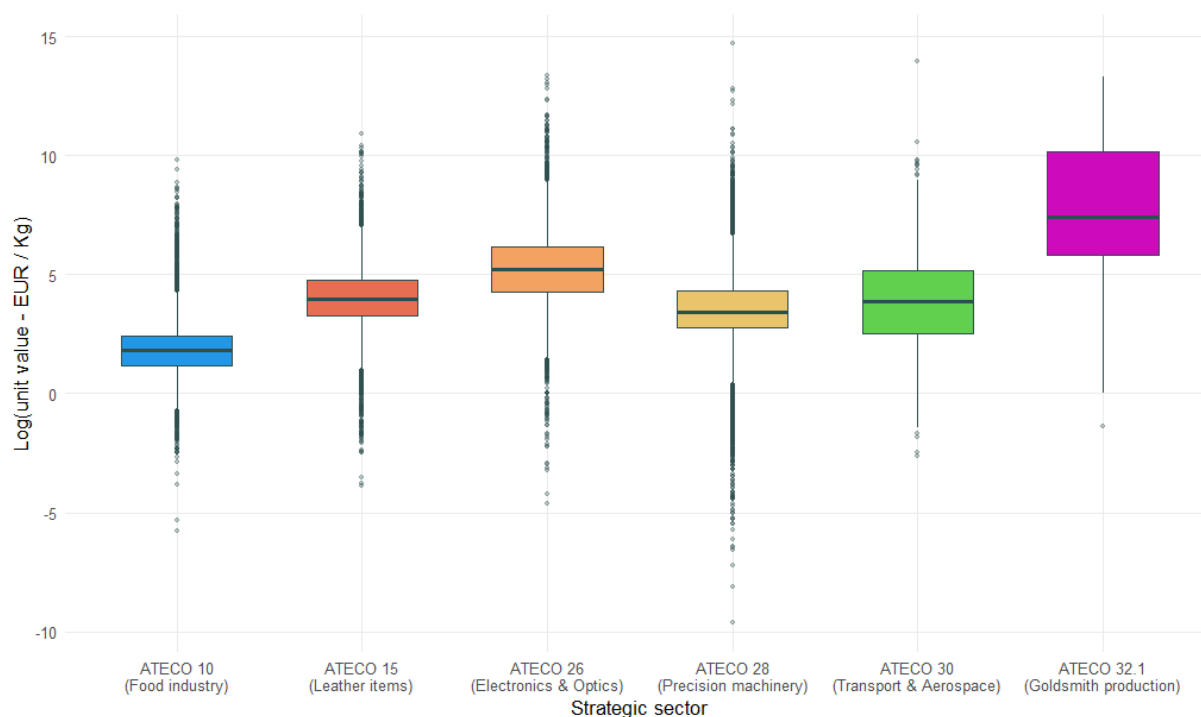


Note. Own elaboration based on ISTAT data (2024).

Evaluating this distribution through an academic lens reveals that analysing aggregate regional export data risks obscuring the unique microeconomic realities driving local industries. Focusing on specific sectors is crucial for deep diving into regional performance for revealing micro-heterogeneity, overcoming the "representative firm" fallacy. As long-tail trade data follows a highly skewed, Pareto-like distribution, treating Veneto as a single industrial entity introduces severe selection bias. ATECO 28 (*Manufacture of machinery and equipment*) operates under a scale-intensive, specialized-supplier configuration characterized by long-term contract enforcement and heavy logistical structures. Conversely, the goldsmith clusters of Vicenza (ATECO 32.1) generate extreme export unit values (€/kg), allowing them to bypass traditional geodetic distance frictions (Alchian-Allen effect). Blending these distinct operational profiles into a single regional average obscure the contrasting strategies deployed by local firms and industrial districts to navigate Global Value Chains. The overwhelming dominance of ATECO 28 and the high qualitative sorting of ATECO 26 (*Manufacture of computers, electronics and optical products; electromedical apparatus, measuring apparatus and clocks*) and ATECO

30 (*Manufacture of other means of transport*) illustrate the structural transformation of the Veneto's industrial districts, moving away from localized, self-contained clusters toward integrated global networks (Corò & Volpe, 2003). The €15.75 Billion generated by precision mechanics cannot be sustained by isolated regional firms. Instead, it relies on highly specialized local networks that interface directly with Global Value Chains. Isolating these advanced sectors allows researchers to evaluate the role of "pivot firms", which act as system integrators that manage international supply networks while anchoring high-value design and production within the home region (Corò & Micelli, 2006). Applying identical econometric assumptions to traditional sectors like food processing (ATECO 10) and scientific frontiers like electronics (ATECO 26) oversimplifies the varied ways industries absorb macroeconomic shocks. High-technology and advanced machinery sectors face rapid product obsolescence, steep entry barriers, and intense R&D requirements (Pavitt, 1984). Analysing these strategic vanguard sectors independently allows for a more precise evaluation of how network centrality measures, institutional risk, and destination-side market traits (GDP and geodetic distance) interact. This targeted approach reveals the distinct mechanisms through which Veneto's advanced manufacturing firms maintain competitive resilience amid global institutional volatility.

Figure 23. Distributional Profiles of Export Unit Values Across the Strategic Sectors



Note. Own elaboration based on ISTAT data (2024).

Figure 23 maps out product quality sophistication and pricing power, leveraging the log of export unit value as a standard empirical proxy (Hallak, 2006). Shifting the focus from absolute value to unit values completely alters the sectoral hierarchy, revealing clear differences in technological intensity and industrial structure. Goldsmith sector (ATECO 32.1) stands completely decoupled from the rest of the manufacturing spectrum, displaying a median log unit value dramatically higher than any other strategic sector. This visual layout confirms the vertical specialization archetype. The Vicenza goldsmith cluster operates on extreme value-to-weight ratios. These firms insulate their margins from geodetic distance frictions by embedding high tacit design capabilities and precious-metal inputs directly into their export goods. Moving across the technology spectrum, a clear gradient emerges matching Lall's (2000) taxonomy. ATECO 26 (Electronics & Optics) occupies a significantly higher pricing power position than the traditional mechanical backbone of ATECO 28. This reflects the science-based technical change regimes (Pavitt, 1984) of advanced optics and electronic components, which command structural price premiums due to rapid product innovation and steep R&D-driven entry barriers. While ATECO 28 (precision machinery) dominated absolute financial weight, its unit-value distribution is compressed and positions lower on the log vertical scale. Crucially, its outlier pattern is highly dense, stretching deep into both the upper and lower limits. The internal composition of ATECO 28 reveals a deeply layered industrial hierarchy. The sector transitions from a high-volume base of standardized component manufacturers—such as fluid dynamics and transmission producers (ATECO 28.1)—to an elite tier of specialized suppliers found in special-purpose machinery (ATECO 28.9). This structural variance explains the immense dispersion observed in our firm-level trade margins: while component sub-contractors scale horizontally through extensive margin expansion across global distribution lines, the specialized systems integrators and industrial robotics builders (ATECO 28.99) command massive unit-value premiums by deploying custom, high-tacit-knowledge engineering solutions directly into premium destination hubs. The boxplot of export unit values of Veneto's firms across the strategic sectors demonstrate why quantile regression architecture is required. Traditional OLS models calculate coefficients based solely on the conditional mean, which would collapse these rich structural variations into a single, unrepresentative point. The visual evidence proves that the determinants of internationalization operate under entirely different structural frameworks depending on the sector.

The empirical results illustrated in the following quantile regression models present a high-resolution micro econometric deconstruction of Veneto's precision machinery sector (ATECO 28) which represents over one-fifth of Veneto's outbound trade flows. By tracking the conditional parameters from $\tau = 0.10$ to $\tau = 0.90$ via a granular 0.05 quantile step, these models expose profound structural non-linearities. The application of Powell's non-parametric Kernel density estimator (1991) to compute the sandwich covariance matrix proves methodologically vital in this context: it provides tight, heteroscedasticity-robust confidence intervals that allow us to definitively identify where coefficients structurally diverge across the firms distribution without the computational latency of traditional resampling such as 'nid' method (non-independent and identical distributed) or 'boot' approach (bootstrap). In the context of quantile regression, the asymptotic covariance matrix of the design parameters $\hat{\beta}(\tau)$ takes a classic "sandwich" form:

$$\Lambda = \tau(1 - \tau)[D_1(\tau)]^{-1}D_0[D_1(\tau)]^{-1} \quad (7)$$

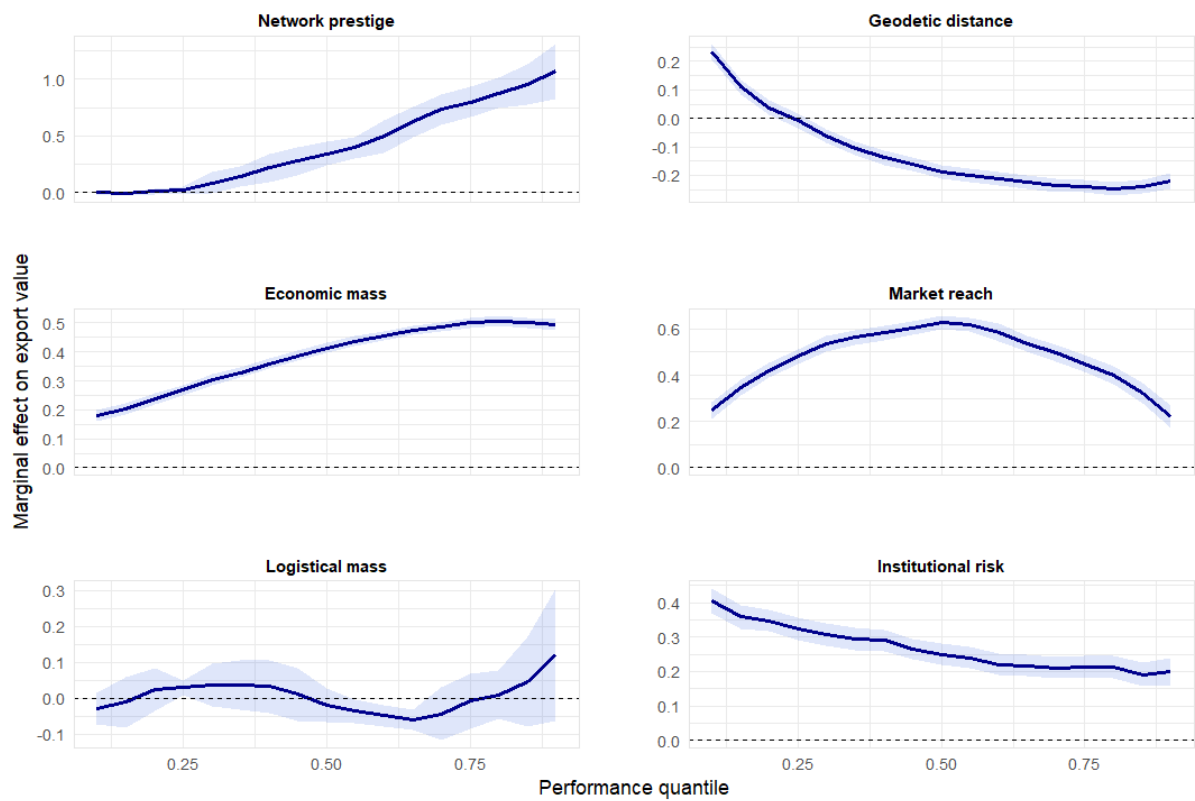
where $D_0 = E[x_i x_i']$ represents the design matrix mass (describes the variation and geometric properties of the covariates). and $D_1(\tau)$ is the structural matrix that uniquely captures the influence of the error terms. $D_1(\tau) = E[f_u(0 | x_i) \cdot x_i x_i']$, where $f_u(0 | x_i)$ represents the conditional density of the residuals evaluated at the τ -th quantile. Because this true density function is unobserved, the empirical challenge lies entirely in estimating $D_1(\tau)$ accurately. Powell's approach bypasses parametric constraints by deploying a non-parametric kernel density function to estimate this localized residual density vector:

$$\widehat{D}_1(\tau) = \frac{1}{nh_n} \sum_{i=1}^n K\left(\frac{\widehat{u}_i(\tau)}{h_n}\right) x_i x_i' \quad (8)$$

where $\widehat{u}_i(\tau) = y_i - x_i' \hat{\beta}(\tau)$ are the fitted quantile residuals. $K(\cdot)$ is a symmetric kernel weighting function that integrates to 1, and h_n is a bandwidth parameter that shrinks to zero as the sample size $n \rightarrow +\infty$. By localized weighting of observations near the τ -th hyperplane via the kernel function, the estimator seamlessly accommodates non-constant error variances across distinct firms' strata. It uses a continuous smoothing kernel across the entire residual vector. This ensures stable, positive variance matrices at any point on

the quantile grid, making it structurally superior for handling tail sparsity. The comparative analysis of Network Prestige (*eigen_std*) in the following quantile models reveals how social capital and systemic interconnectedness shift from a driver of export value to an anchor of pricing premiums. The contrasting layouts of Geodetic Distance (*log (Distance_km)*) and Market Reach (*out_degree_std*) map out the distinct boundaries between expanding a firm's geographic footprint versus intensifying foreign market transactions. The behavior of the Institutional Risk Index (first principal component of PCA, standardized) across the quantile models offers a sophisticated look at how regional enterprises insulate themselves from destination-side political and financial fragility.

Figure 24. Quantile Regression With Export Value as Response Variable

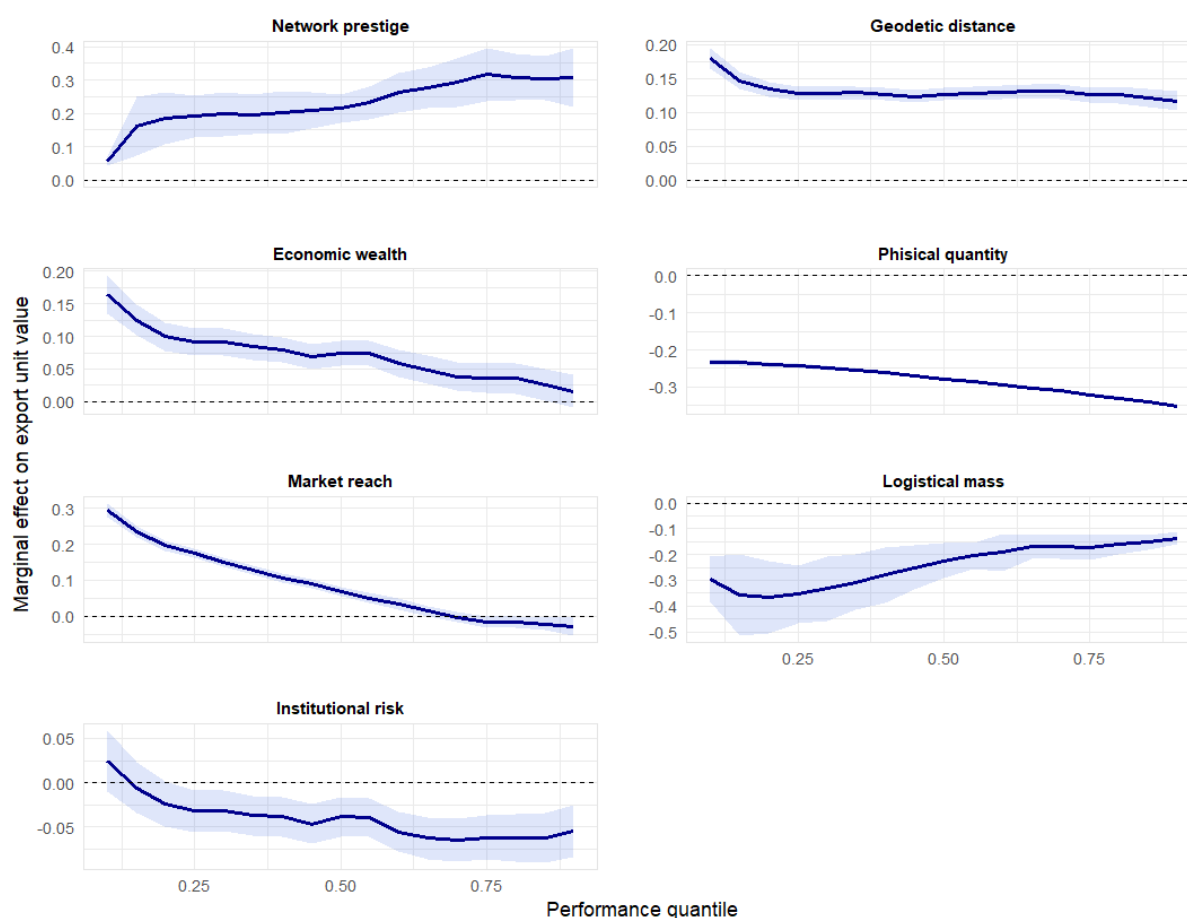


Note. Own elaboration based on ISTAT, SACE and WDI data (2024).

In the first quantile model (Figure 24), the marginal effect of Network Prestige is statistically negligible for the lowest-performing tier ($\tau \leq 0.20$), but scales sharply and monotonically, maximizing its impact at the 90th percentile ($\beta \approx 1.05$). This indicates that for Veneto's mechanical "superstar exporters," structural centrality within Global Value Chains acts as a massive force multiplier for absolute export volume. Geodetic Distance exhibits a classic parabolic decline. It is positively signed at the lowest deciles (reflecting small firms operating in contiguous, high-value border markets) but plunges

deep into negative elasticity ($\beta \approx -0.25$) for the upper quantiles. Concurrently, Market Reach reveals an inverted-U trajectory, peaking near the median ($\tau = 0.50$). This proves that mid-tier machinery firms expand their aggregate export volume by maximizing geographic diversification (horizontal scaling). For the top-tier superstars ($\tau \geq 0.75$), the marginal return of entering new destination markets drops sharply. Institutional Risk Index carries a universally positive coefficient that systematically decays as we move rightward along the distribution. Lower-tier exporters ($\tau = 0.10$) show an extreme positive sensitivity ($\beta \approx 0.40$), indicating their volumes are highly dependent on entering volatile, high-risk/high-yield developing markets. As firm performance scales up toward the superstar vanguard, this elasticity cuts in half, settling at a highly stable $\beta \approx 0.20$, demonstrating severe structural resilience and volume decoupling from target country volatility. The Economic Mass ($\log(GDP_Eur)$) scales upward, showing that larger destination markets disproportionately pull absolute volumes from larger and best-performing firms.

Figure 25. Quantile Regression With Export Unit Value as Response Variable



Note. Own elaboration based on ISTAT, SACE and WDI data (2024).

In the second quantile model (Figure 25), Network Prestige behaviour transitions into a structural step-function. The parameter jumps immediately at $\tau = 0.15$ to $\beta \approx 0.20$ and remains positive and statistically significant up to the highest deciles ($\beta \approx 0.30$). The elasticity of Geodetic Distance is uniformly positive ($\beta \approx 0.12$) across all performance quantiles. This is a striking empirical manifestation of the Alchian-Allen effect (or quality-sorting hypothesis): as shipping and logistical distances lengthen, firms selectively ship higher-quality, more expensive machine configurations to absorb fixed transportation costs. Interestingly, the marginal effect of Market Reach on unit values is sharply negative for high-performing firms, dipping below zero at $\tau \geq 0.70$. This confirms that top-tier mechanical firms maximize quality premiums by pursuing a hyper-selective, highly concentrated vertical market strategy rather than diversifying across too many destinations. Institutional Risk registers an entirely different regime. For marginal exporters ($\tau \leq 0.15$), the effect is positive, but for the middle and upper-tier mechanical vanguards ($\tau \geq 0.25$), the parameter turns stably negative ($\beta \approx -0.04$). This uncovers a harsh economic reality for Veneto's premium manufacturers: when exporting to fragile institutional environments, even high-performing firms face downward export value compression. They are forced to downgrade their technical specifications. Economic Wealth ($\log(GDP_per_capita)$) exhibits a downward trend, showing its strongest premium extraction capacity ($\beta \approx 0.16$) among lower-performing firms, while superstar pricing power remains quite insulated from destination income levels. Physical Quantity ($\log(total_qua)$) displays a flat, structurally negative elasticity ($\beta \approx -0.25$ to -0.35) across all quantiles. This confirms a universal volume-discount regime across the machinery sector, proving that as physical bulk increases, unit values drop. Conversely, Logistical Mass ($out_strength_std$) displays a negative elasticity at the lower tail that recovers toward zero for top-tier firms, demonstrating that elite manufacturers possess the sophisticated logistical infrastructure necessary to neutralize bulk shipping penalties on their premium equipment.

By utilizing an inter-quantile regression framework robustly mapped via Kernel-sandwich estimators, this comparative analysis proves that Veneto's precision machinery sector operates under dual economic regimes. Absolute export value is driven by horizontal market diversification and network centrality concentrated heavily within a fractional elite of superstar consolidators (Melitz, 2003). Conversely, qualitative pricing

power behaves as a democratized asset where network prestige delivers immediate premiums across the board, and distance serves as a quality-sorting mechanism. These findings mathematically invalidate the representative-firm assumption, pioneered by economist Alfred Marshall in the late 19th century, demonstrating that a firm's operational position on the performance grid fundamentally dictates its capacity to handle geographic frictions and institutional risk. The empirical outputs presented in Figure 26 and Figure 27 provide a definitive, non-parametric validation of behavioural asymmetry across the distribution of Veneto's precision machinery sector. To establish a mathematically rigorous validation of the heterogeneous firm paradigm pioneered by Melitz (2003), this study implements an inter-quantile Wald test comparing the extreme deciles of the conditional export distribution ($\tau = 0.10$ and $\tau = 0.90$). By executing the Robust Wald Test (ANOVA), setting `joint = FALSE` tests each variable's slope equality independently across the quantiles, and `se = "ker"` ensures high-speed, robust covariance calculation. While traditional empirical literature frequently relies on inter-quartile ranges ($\tau = 0.25$ and $\tau = 0.75$), such standard thresholds lack the resolution required to isolate the actual "superstar exporter" vanguard highlighted by Freund and Pierola (2015). Testing for parameter structural equality at the outer boundaries of the firm distribution ensures an accurate contrast between the marginal, risk-averse SME population ($\tau = 0.10$) and the hyper-connected, high-complexity systems integrators ($\tau = 0.90$) driving Veneto's precision machinery sector (Corò & Micelli, 2006). Potential variance inflation inherent to tail estimations is mitigated via the application of Powell's non-parametric kernel density estimator, ensuring robust inference under severe heteroscedasticity.

Figure 26. Inter-Quantile Wald Test on the Quantile Model With Export Value

Quantile Regression Analysis of Deviance Table

Model: `log(total_val) ~ Risk_Index_Standardized + log(Distance_km) + log(GDP_Eur) + out_degree_std + out_strength_std + eigen_std`
 Tests of Equality of Distinct Slopes: `tau in { 0.1 0.9 }`

	Df	Resid Df	F value	Pr(>F)	
Risk_Index_Standardized	1	102459	70.9004	<2e-16	***
log(Distance_km)	1	102459	568.3720	<2e-16	***
log(GDP_Eur)	1	102459	592.3606	<2e-16	***
out_degree_std	1	102459	1.0011	0.3171	
out_strength_std	1	102459	2.5981	0.1070	
eigen_std	1	102459	73.8569	<2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Note. Own elaboration based on ISTAT, SACE and WDI data (2024).

The analysis of Deviance Table for absolute export value reveals highly significant structural slope mutations across the extreme cohorts ($N_{df_resid} = 102,459$). The formal rejection of the null hypothesis ($H_0: \beta_{0.10} = \beta_{0.90}$) across specific covariates yields critical macro- and micro-level insights. Both Geographical Distance and Economic Mass of the foreign country display massive F-statistics (568.37 and 592.36, respectively) with p-values approaching zero ($p < 2 \times 10^{-16}$). This provides robust empirical proof that traditional gravity forces do not affect marginal exporters and superstar vanguards uniformly. As firms scale up, their elasticities to spatial distance and destination size shift significantly, reflecting the varying capacities of these cohorts to overcome the sunk costs of internationalization. The slope equality test of Risk Index Standardized is soundly rejected ($F = 70.90, p < 2 \times 10^{-16}$) because destination-side institutional fragility acts as a highly non-linear vector. The marginal SME cohort at $\tau = 0.10$ faces a completely different structural exposure to institutional volatility compared to the resilient system integrators at $\tau = 0.90$. Intriguingly, the Extensive Margin and Logistical Mass fail to reject the null hypothesis of slope equality, displaying statistically non-significant F-values (1.00 and 2.60, respectively). However, Network Prestige strongly rejects parameter stability ($F = 73.86, p < 2 \times 10^{-16}$). This suggests that while expanding Market Reach or physical volume exerts a statistically parallel effect across the entire firm spectrum, relational prestige and integration into core international hubs offer exponentially higher marginal returns exclusive to the superstar elite.

Figure 27. Inter-Quantile Wald Test on the Quantile Model With Export Unit Value

Quantile Regression Analysis of Deviance Table

Model: $\log(\text{unit_val}) \sim \log(\text{total_qua}) + \text{Risk_Index_Standardized} + \log(\text{Distance_km}) + \log(\text{GDP_per_capita}) + \text{out_degree_std} + \text{out_strength_std} + \text{eigen_std}$
 Tests of Equality of Distinct Slopes: tau in { 0.1 0.9 }

	Df	Resid	Df	F value	Pr(>F)
$\log(\text{total_qua})$	1	102459	678.097	< 2.2e-16	***
$\text{Risk_Index_Standardized}$	1	102459	13.087	0.0002975	***
$\log(\text{Distance_km})$	1	102459	38.317	6.037e-10	***
$\log(\text{GDP_per_capita})$	1	102459	63.680	1.554e-15	***
out_degree_std	1	102459	437.097	< 2.2e-16	***
out_strength_std	1	102459	12.602	0.0003855	***
eigen_std	1	102459	31.680	1.823e-08	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Note. Own elaboration based on ISTAT, SACE and WDI data (2024).

When shifting the analytical lens to qualitative pricing power and quality sorting via the

log of export unit value (Figure 27), every single covariate strongly rejects the null hypothesis of slope equality across the extreme deciles. This comprehensive rejection demonstrates that the pricing architectures of marginal firms and superstar exporters are governed by entirely distinct economic regimes. The scale elasticity parameter displays a massive structural divergence ($F = 678.10$, $p < 2.2 \times 10^{-16}$). This confirms that the volume-discount penalties and price-compression mechanisms associated with large physical trade flows operate with highly asymmetric intensities between small suppliers and dominant aggregators. The distance coefficient shifts significantly between the tails ($F = 38.32$, $p = 6.037 \times 10^{-10}$). This indicates that the "shipping the good apples out" mechanism is mediated by firm-level performance. True superstar firms leverage distance as a deliberate quality-sorting vehicle far more aggressively than marginal operators. In stark contrast to the volume model, all three network centrality metrics register highly significant differences in slope equality. The Market Reach (*out_degree_std*) displays an exceptionally high F-statistic (437.10 , $p < 2.2 \times 10^{-16}$). This provides definitive evidence that the structural trade-off between expanding market diversity (horizontal scaling) and maximizing localized quality premiums (vertical scaling) changes radically as a firm approaches the elite vanguard. These findings confirm that the microeconomic mechanisms of internationalization are highly non-linear. The localized, tacit knowledge networks historically anchored within Veneto's specialized supplier districts (Becattini, 1990) have evolved into highly polarized systems. Within these ecosystems, marginal SMEs and hyper-connected systems integrators operate on entirely separate strategic planes to manage geographic friction, leverage network assets, and absorb institutional risk across Global Value Chains.

5.3 Sectoral Deep Dive: Decomposition of Internationalization Regimes

The summary matrix of descriptive statistics presented in Table 5 offers a high-resolution, empirical micro-diagnostic of the trade profiles across Veneto's primary manufacturing sectors. Econometrically, a comprehensive analysis of export intensity—defined as the ratio of foreign sales to total corporate turnover—would have been highly suitable in this context, as it isolates a firm's structural dependency on international markets relative to its domestic base. However, the absence of relevant balance-sheet data (such as total revenues) does not allow for this specific computation. To circumvent this data limitation

while preserving a rigorous evaluation of internationalization strategies, this study instead decomposes trade performance by analyzing the Extensive Margin (*out_degree*) with respect to the Intensive Margin (*total_val/out_degree*)—defined as the specific export value extracted per foreign country reached.

Table 5. Summary Statistics Related to Firm-Level Intensive Margin

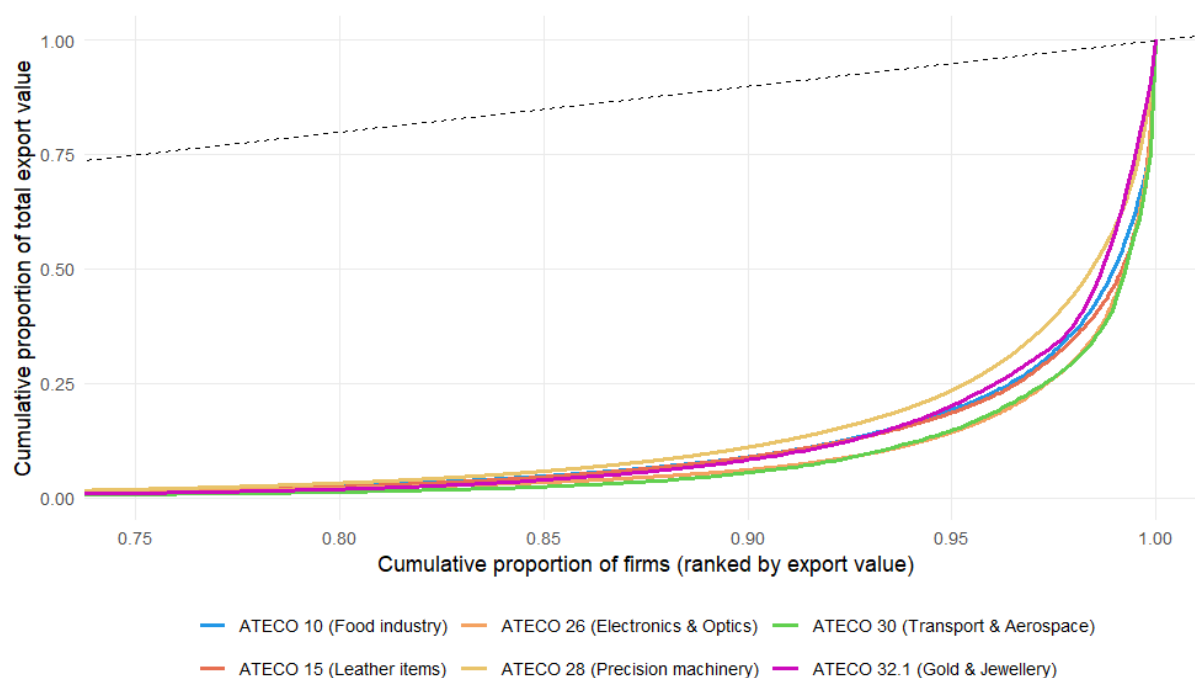
▲	Sector_Label	Active_Firms	Mean_Intensive_Margin	Median_Intensive_Margin	Max_Intensive_Margin	SD_Intensive_Margin
1	ATECO 10 (Food industry)	1805	214417.46	16687.80	55288773	1500560.7
2	ATECO 15 (Leather items)	1932	255052.42	7680.00	107119413	2955412.6
3	ATECO 26 (Electronics & optics)	2543	36987.79	1640.00	6020181	211217.9
4	ATECO 28 (Precision machinery)	6742	128564.75	12805.48	33393399	602374.4
5	ATECO 30 (Transport & Aerospace)	748	172697.52	2297.40	46454829	1791687.6
6	ATECO 32.1 (Goldsmith production)	728	418719.26	8530.50	138355625	5203132.8

Note. Own elaboration based on ISTAT, SACE and WDI data (2024).

The defining characteristic across all six industrial domains is the severe, systematic divergence between the conditional mean and the conditional median. In a homogenous distribution of exporters, these metrics would converge. Here, the mean exceeds the median by multiple orders of magnitude in all the strategic sectors. This structural signature mathematically demonstrates that the typical, baseline SME exporter operates on a modest scale, serving foreign markets at a minimal intensity. The inflated means are entirely pulled upward by the far right-hand tail of the distribution (represented by the *Max_Intensive_Margin* column) where single "superstar" corporate entities generate individual market intensities exceeding €138 Million (ATECO 32.1) or €107 Million (ATECO 15). This extreme polarization is further reflected in the exceptionally high standard deviations (*SD_Intensive_Margin*), proving that intra-sectoral inequality is a systemic trait of the regional export base rather than a statistical anomaly. When evaluating the cross-sectoral dynamics, the metrics uncover structurally distinct strategies utilized by firms to scale up within Global Value Chains. As the data implies, the paths to achieving "superstar" status are fundamentally differentiated by industry characteristics: a firm scaling up in ATECO 28 focuses primarily on horizontal volume expansion (expanding its extensive margin across foreign countries). Meanwhile, a firm scaling up in ATECO 32.1 relies on vertical quality escalation, maximizing its intensive margin focus to capture high price premiums within selective, high-income destination hubs. This dichotomy is clearly reflected in the structural metrics of the two sectors:

- precision machinery sector (ATECO 28) contains by far the largest population of internationalized actors (6,742 active firms) and displays the highest median intensive margin (€12,805.48) alongside a relatively contained standard deviation (€602,374.4) relative to its mean (€128,564.75). This uniform upward shift in the median indicates a resilient and mature ecosystem. Within Veneto's specialized mechanical supplier districts, engineering capabilities are distributed across a broad mid-tier SME network that scales horizontally by sequentially absorbing the sunk costs of entering multiple new geographic markets.
- Goldsmith sector (ATECO 32.1) operates with a compressed population of operators (728 firms) and generates the highest mean intensive margin in the entire dataset (€418,719.26). Because jewellery and precious metal processing carry an exceptional ratio of monetary value to physical weight, these firms focus on high unit-value sophistication. They insulate their margins from geodetic distance frictions by embedding high tacit design capabilities into their export goods, extracting extreme value depths out of a selective, high-income destination markets.

Figure 28. Lorenz Curves: Signs of Market Concentration in the Strategic Sectors



Note. Own elaboration based on ISTAT, SACE and WDI data (2024).

The mathematical evaluation of export concentration via Gini coefficients and Lorenz curves (Figure 28) provides an uncompromised validation of the heterogeneous firm

paradigm within Veneto's industrial engine. Across all six strategic sectors, the empirical Lorenz curves display a severe deviation from the line of perfect equality, tracing out a Pareto-like concentration signature characteristic of systemic superstar architectures. The data demonstrates that instead of a distributed, egalitarian export base, Veneto's strategic sectors are highly asymmetric. To evaluate the inequalities shown in the Lorenz Curves (where the empirical distributions deviate significantly from the 45-degree dashed line of perfect equality), we compute the Gini Coefficients (Table 6). Mathematically, for a given sector with n firms, where y_i represents the export value of firm i indexed in non-decreasing order ($y_1 \leq y_2 \leq \dots \leq y_n$), the Gini coefficient is formalized as:

$$G = \frac{\sum_{i=1}^n (2i - n - 1) y_i}{n \sum_{i=1}^n y_i} \quad (9)$$

Alternatively, using the geometric properties of the Lorenz Curve plot canvas, the coefficient can be defined as:

$$G = \frac{A}{A + B} = 1 - 2 \int_0^1 L(p) dp \quad (10)$$

where A represents the area of inequality between the diagonal line of perfect equality and the empirical Lorenz curve $L(p)$, and B represents the entire area under the Lorenz curve. A coefficient approaching 1 denotes absolute, monopolistic concentration, a boundary that Veneto's strategic sectors closely approach.

Table 6. Gini Coefficient (Market Concentration) for Each Strategic Sector

	Sector_Label	Gini_Coefficient
1	ATECO 10 (Food industry)	0.9345670
2	ATECO 15 (Leather items)	0.9370604
3	ATECO 26 (Electronics & Optics)	0.9452486
4	ATECO 28 (Precision machinery)	0.9201854
5	ATECO 30 (Transport & Aerospace)	0.9503105
6	ATECO 32.1 (Gold & Jewellery)	0.9337558

Note. Own elaboration based on ISTAT, SACE and WDI data (2024).

Table 7. Market Shares of the Firms That Occupy the Top Percentiles of Population

	Sector_Label	Top_10pct	Top_5pct	Top_1pct
1	ATECO 10 (Food industry)	91.05193	80.72468	49.60169
2	ATECO 15 (Leather items)	91.08238	81.28797	52.98510
3	ATECO 26 (Electronics & Optics)	93.84990	85.65581	55.66453
4	ATECO 28 (Precision machinery)	88.93684	76.51072	40.73913
5	ATECO 30 (Transport & Aerospace)	94.41749	85.03674	55.43947
6	ATECO 32.1 (Gold & Jewellery)	91.59084	79.71598	41.06428

Note. Own elaboration based on ISTAT, SACE and WDI data (2024).

The empirical calculations across the strategic sectors (Table 6 and Table 7) reveal an exceptionally high concentration profiles, with all Gini indices clustering in an upper band between 0.920 and 0.950:

- ATECO 30 (Transport & Aerospace) exhibits the highest systemic concentration in the dataset, with a Gini coefficient of 0.950311. The top 10% of firms control 94.42% of the sector's total export value, while the top 1% alone controls an astonishing 55.44%. This structural profile confirms a strict science-based, high-sunk-cost regime, where international operations are restricted to a tiny group of highly capitalized systems integrators.
- ATECO 26 (Electronics & Optics) registers a Gini coefficient of 0.945249, with its top 5% elite controlling 85.66% of outbound trade value. This profile matches a segmented Global Value Chain layout. A massive fringe of localized sub-contractors handles low-value components, while a small group of highly integrated corporate hubs manages the final high-value export vectors.
- ATECO 28 (Precision Machinery) displays the lowest—though still highly unequal—concentration signature, with a Gini of 0.920185. Its top 5% controls 76.51% of trade value, and its top 1% controls 40.74%. This unique profile matches the specialized-supplier district networks of the Veneto region. The path-dependent engineering capabilities of the region are mature enough to allow a slightly broader base of mid-tier SMEs to maintain a direct international market presence alongside the sector's superstar entities.
- ATECO 32.1 (Gold & Jewellery) carries a Gini of 0.933756, with the top 10% commanding 91.59% of the market. This reveals that despite the artisan legacy of the Vicenza cluster, the commercialization and final shipping of premium-quality jewellery are highly aggregated around a few key global distributors.

The findings mathematically demonstrate that a uniform linear regression would fail to capture these distinct structural realities. The forces guiding international trade—such as geographic distance friction or destination-side institutional risk—operate on completely different scales for the marginal 90% of firms compared to the highly concentrated 1% to 5% corporate elite that commands most of the Veneto's export mass. The elevated levels of concentration in the strategic sectors proves that Veneto's competitive resilience in Global Value Chains is fundamentally reliant on the stability and relational capital of its elite industrial vanguard. This extreme polarization introduces distinct macroeconomic advantages and structural vulnerabilities that define the competitive resilience of Veneto's export base. Entering foreign markets, navigating complex regulatory compliance, and mitigating destination institutional risk require heavy investments. Only highly capitalized "superstar exporters" possess the resources necessary to absorb these fixed costs. Large elite firms act as "pivot firms", serving as critical hubs that connect localized, fragmented SME sub-suppliers to international buyers (Corò & Micelli, 2006). They secure global contracts that small, isolated firms could never capture independently. Concentration allows top-tier firms to accumulate the capital reserves necessary to finance long-term R&D, advanced testing facilities, and automation technologies. This continuous investment drives industrial upgrading and reinforces the quality premiums of the region's medium and high-technology sectors. Because the top 1% of firms routinely control over 40% to 55% of total sectoral export value, the aggregate trade performance of the entire Veneto region is highly dependent on the financial health of a few corporate actors. If an idiosyncratic shock (e.g., supply chain blockages, bankruptcy, or strategic reallocation) hits a single superstar firm, it can trigger a severe contraction in the region's overall export figures. Severe concentration can indicate a hollowing out of the traditional industrial district fabric. If the gap between the marginal SMEs and the superstar vanguard becomes too wide, middle-tier firms disappear, breaking down the localized, collaborative learning loops that historically drove the resilience of the North-Eastern Italian economy.

5.4 High-Dimensional PPML Gravity Estimations

To uncover the structural determinants governing the export value of Veneto's strategic manufacturing backbone, this study transitions from descriptive micro-data mapping to

latest formal econometric modelling. Standard linear trade regressions frequently suffer from severe estimation biases due to the presence of zero-trade flows and non-constant error variances. To systematically overcome these limitations, we implement the Poisson Pseudo-Maximum Likelihood (PPML) estimator pioneered by Santos Silva and Tenreyro (2006). By evaluating the dependent trade variables in their native multiplicative levels rather than relying on log-linearized transformations, the PPML framework preserves zero-valued bilateral matrices and ensures complete consistency even under severe heteroscedastic conditions. To establish full econometric identification, the empirical specifications engineered for the selected strategic sectors must carefully navigate the dual requirements of accounting for unobserved heterogeneity and ensuring robust statistical inference. Within this structural model, Fixed Effects (FE) operate as specialized categorical intercept shifters. Their primary methodological objective is to absorb and neutralize unobserved heterogeneity—the unmeasured, hidden characteristics latent within granular trade data that fundamentally impact international trade volumes but escape simple indexation. In the absence of a robust fixed effects structure, these unobserved traits end up trapped within the model's composite error term. If these omitted characteristics correlate with the explanatory variables—such as destination-side institutional risk or firm-level network centrality measures—the model suffers from severe omitted variable bias (endogeneity), rendering the estimated coefficients mathematically incorrect and invalidating any causal interpretation. In enterprise-level trade micro-data, this endogeneity typically manifests through two primary channels. First, firm fixed effects (*firm_ID*) are required to absorb everything unique, time-invariant, and unchanging about an individual enterprise, such as its baseline productivity, idiosyncratic managerial talent, or global brand reputation. Second, foreign country fixed effects (*Paese*) are utilized to capture everything unique about a specific destination market, effectively neutralizing unobserved macro-frictions like cultural open-door policies, underlying language barriers, or historical bilateral trade agreements.

While the configuration of fixed effects alters the conditional mean coefficients, clustered standard errors adjust the model's standard errors and p-values to safeguard statistical confidence. Traditional ordinary least squares regressions operate under the assumption that every observation within the sample matrix is completely independent of every

other. In enterprise-level micro-data, this baseline assumption is fundamentally unrealistic. For instance, firms operating within the same geographic province (e.g., Vicenza) inevitably share identical or similar infrastructure, localized logistics pipelines, and interconnected banking networks, meaning their cross-border behaviours are inherently correlated. Similarly, distinct trade flows directed toward the same foreign country (e.g., Germany) are bound by the same macroeconomic shocks, institutional regulatory updates, and currency fluctuations, regardless of the individual firm executing the export movement. Failing to account for these intra-group dependencies causes statistical software to assume it possesses significantly more unique, independent information than the dataset actually contains. This artificially deflates the standard errors, inflating the model's precision and causing random noise to appear highly statistically significant ($p\text{-value} < 0.01$). To counteract this variance distortion, this study specifies multi-way clustered standard errors¹² across the province where the firm is located and the foreign country to which the goods were exported. This configuration forces the estimation algorithm to allow observations within the same geographic province of origin or destination country to be correlated with one another, generating structurally honest standard errors that secure the validity of our inductive inferences.

The core structural challenge of this thesis lies in the simultaneous identification of firm-specific network metrics and macro-gravity parameters. In modern gravity trade theory (Anderson & van Wincoop, 2003), a firm's propensity to export depends heavily on Multilateral Resistance Terms (MRTs). This framework dictates that an enterprise does not evaluate a target market in isolation; rather, it decides to export to a specific country (e.g., Germany) based on how easy or difficult it is to access that country relative to all other alternative global markets simultaneously, conditioned by its local operational environment. While standard gravity literature recommends the inclusion of strict firm fixed effects (*firm_ID*) to fully absorb these origin-side MRTs, doing so drops any explanatory variables that do not vary across destinations for a given firm. Because our primary firm-level independent variables—namely Network Prestige (*eigen_std*), Market Reach (*out_degree_std*), and Logistical Mass (*out_strength_std*)—are structural attributes assigned statically to each enterprise, placing *firm_ID* in the fixed effects dimension triggers a perfect collinearity trap, making it mathematically impossible to calculate their

¹² Through cluster = c("provinci", "Paese"), specified in the summary of the econometric models

respective coefficients. To resolve this identification barrier, this study implements a rigorous econometric compromise. By utilizing an interacted structural fixed effect composed of the firm's geographic province of origin and the strategic sector to which the trade flows belong (*provinci^ateco_prefix*), the models successfully account for origin-side multilateral resistance terms without absorbing the independent variance of our network metrics. In the practical context of North-Eastern Italy's economic architecture, this combination makes profound economic sense. Veneto's manufacturing engine is historically rooted in specialized industrial districts, meaning that a mechanical components firm in Vicenza operates within a completely different industrial ecosystem, supply-chain network, and labour pool than a textile manufacturer located in Treviso. By locking the combined interaction (*provinci^ateco_prefix*) into the fixed effects dimension, the model controls for the shared background advantages of these localized clusters: their specialized sub-supplier networks, belonging to the same strategic sector, and shared pools of tacit engineering knowledge.

Figure 29. PPML Model 1 With *ateco_prefix* as Fixed Effects

```
GLM estimation, family = poisson, Dep. Var.: total_val
Observations: 99,390
Fixed-effects: ateco_prefix: 6
Standard-errors: Clustered (provinci & Paese)

      Estimate Std. Error   z value   Pr(>|z|)
log(Distance_km)    -0.298911    0.086688   -3.44812  5.6450e-04 ***
log(GDP_Eur)         0.485643    0.042286   11.48484  < 2.2e-16 ***
Risk_Index_Standardized 0.168570    0.059158    2.84950  4.3788e-03 **
out_degree_std       0.410419    0.056634    7.24689  4.2644e-13 ***
out_strength_std     0.143847    0.040391    3.56132  3.6899e-04 ***
eigen_std            0.059112    0.048585    1.21669  2.2372e-01
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Log-Likelihood: -6.357e+10   Adj. Pseudo R2: 0.177877
               BIC: 1.271e+11   Squared Cor.: 0.019644
```

Note. Own elaboration based on ISTAT, SACE and WDI data (2024).

Figure 30. PPML Model 2 With Combined Fixed Effects (*province^ateco_prefix*)

```
GLM estimation, family = poisson, Dep. Var.: total_val
Observations: 99,390
Fixed-effects: provinci^ateco_prefix: 42
Standard-errors: Clustered (provinci & Paese)

      Estimate Std. Error   z value   Pr(>|z|)
log(Distance_km)    -0.308799    0.088254   -3.49899  4.6702e-04 ***
log(GDP_Eur)         0.490867    0.043494   11.28573  < 2.2e-16 ***
Risk_Index_Standardized 0.165541    0.056232    2.94388  3.2413e-03 **
out_degree_std       0.428997    0.056728    7.56234  3.9589e-14 ***
out_strength_std     0.107381    0.050471    2.12759  3.3371e-02 *
eigen_std            0.136981    0.069510    1.97066  4.8763e-02 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Log-Likelihood: -6.082e+10   Adj. Pseudo R2: 0.213529
               BIC: 1.216e+11   Squared Cor.: 0.031735
```

Note. Own elaboration based on ISTAT, SACE and WDI data (2024).

A systematic review of the covariates across both specifications (Figure 29 and Figure 30) reveals robust structural stability, though the magnitude and statistical precision of the coefficients undergo vital mutations as the fixed effects dimension narrows:

- **Geodetic Distance:** across both specifications, geographical friction exhibits a highly stable, negative elasticity. In Model 1, the coefficient is -0.299, shifting slightly to -0.309 in Model 2, maintaining strict statistical significance ($p < 0.001$). This confirms that even within highly specialized strategic supply chains, a 1% increase in distance imposes a robust $\approx 0.31\%$ structural transport penalty on trade flows.
- **Economic Mass:** destination market size functions as the primary pull factor for regional trade mass, displaying highly significant, positive elasticities of 0.486 in Model 1 and 0.491 in Model 2 ($p < 2.2 \times 10^{-16}$). This indicates that a 1% expansion in target market economic capacity scales Veneto's export goods (belonging to the selected strategic sectors) by approximately 0.49%.
- **Institutional Risk:** the institutional risk parameter captures a positive, statistically significant elasticity across both specifications (0.169 in Model 1 and 0.166 in Model 2 with $p < 0.01$). While intuitive economic logic might predict a negative relationship, this positive coefficient is highly representative of Veneto's specialized manufacturing sectors. It reveals that regional firms actively exploit high-risk/high-yield developing economies, where localized infrastructural gaps trigger a surge in demand for direct capital goods imports.
- **Market Reach:** the standardized number of foreign target markets reached exerts an immense upward pressure on total export value across both regimes, scaling from 0.410 to 0.429 ($p < 0.001$). Broad market diversification remains an absolute prerequisite for volume accumulation.
- **Logistical Mass:** the standardized strength of a firm's network distribution remains positively signed but exhibits a sharp degradation in magnitude and significance. It drops from a highly significant 0.144 ($p < 0.001$) under macro fixed effects to a marginal 0.107 ($p < 0.05$) under localized cluster conditioning.
- **Network Prestige:** the behavior of eigenvector centrality provides the most compelling econometric justification for the structural transition between the two specifications. In Model 1, when conditioning only on broad strategic sector categories (*ateco_prefix*), network prestige is completely statistically insignificant ($\beta = 0.059$, $p = 0.223$). However, in Model 2, when controlling for the exact localized peer cluster, the

coefficient more than doubles to 0.137 and becomes statistically significant ($\beta = 0.137, p < 0.05$).

Model 2 is unequivocally the superior econometric specification for this study. The transition from a broad macro-sector dummy ($ateco_prefix = c("10", "15", "26", "28", "30", "32.1")$) to an interacted regional-product fixed effect category ($provinci^ateco_prefix$) represents a vital step-function change in model identification. By intersecting the geographic point of origin (the province where is located the firm) with the specific strategic industry, Model 2 successfully absorbs unobserved local industrial capabilities, district-level structural infrastructure, and the shared background assets unique to North-Eastern Italian industrial districts. The mathematical parameters strongly support this selection. Model 2 triggers a massive upward shift in the Log-Likelihood, which flows directly into a substantial reduction in the Bayesian Information Criterion (BIC) from 1.271×10^{11} down to 1.216×10^{11} . Because the BIC heavily penalizes overfitting and the introduction of unnecessary parameters, this drop proves that moving to 42 interacted fixed-effects groups adds genuine, non-redundant explanatory value. Concurrently, the Adjusted Pseudo R^2 increases from 17.79% to 21.35%. Most crucially, the activation of Network Prestige ($eigen_std$) from statistical insignificance in Model 1 to a robust, positive driver in Model 2 confirms that Model 1 suffered from severe omitted variable bias. In Model 1, the shared structural advantages of specialized clusters (e.g., being a mechanical firm in the highly optimized network fabric of Vicenza vs. an isolated operator elsewhere) were left in the error term, confounding the network prestige coefficient. By capturing these district-level background advantages through the $provinci^ateco_prefix$ intercept shifter, Model 2 isolates the true, independent return of a firm's individual network architecture.

5.5 The Impact of an Institutional Risk Shock on Regional Exports

To transition from the conditional mean parameter estimates to a dynamic evaluation of structural resilience across Veneto's industrial backbone, this section implements an explicit macroeconomic simulation framework. Within contemporary international trade networks, evaluating baseline elasticities is often insufficient for predicting cluster survivability under conditions of heightened geopolitical and sovereign volatility. Consequently, conducting a formal stress test becomes methodologically imperative. By

subjecting our micro-level data to simulated institutional tension, we move beyond static descriptions of trade flows to isolate the non-linear value-destruction paths across different firm size and performance strata. This experimental setup evaluates this systemic vulnerability by engineering a direct comparison between an unshocked baseline scenario—representing the conditional expectations of observed trade levels of 2024—and a severe shocked scenario defined by a +2 standard deviation escalation in destination-side Institutional Risk Index which contains information about credit and political risk. Because the underlying Poisson Pseudo-Maximum Likelihood (PPML) gravity engine models bilateral trade value in native levels via an exponential link function, a linear adjustment to our standardized foreign country risk metric scales up absolute trade outcomes multiplicatively. It translates a raw policy shock into a concrete metric of sector-specific revenue capital at risk. However, rendering this predictive transformation visually across a multi-dimensional field of macro-gravity invariants and corporate network attributes presents a profound geometric challenge. To establish a clean, readable coordinate space, we deploy the mathematical foundation of Principal Component Analysis (PCA) as an orthogonal dimensionality reduction tool. PCA takes our core structural control variables—including spatial distance, GDP of the foreign country, and our three standardized firm-level network topology metrics—and compresses their variance into two principal components, PC_1 and PC_2 . Because the underlying data matrix is pre-standardized via Z-score scaling ($\mu = 0, \sigma = 1$), the transformation neutralizes unit-of-measurement discrepancies while fully preserving the true linear variance gaps separating diverse market destinations and firm capabilities. The resulting horizontal plane cleanly isolates variations in *Firm & Network Scale Capacity* (PC_1) and *Spatial Friction & Target Market Scale* (PC_2), creating an invariant visual testing landscape.

Table 8. Importance Table Resulting From Principal Component Analysis

	PC1	PC2	PC3	PC4	PC5
Standard deviation	1.289935	1.123543	1.019284	0.8134113	0.6108523
Proportion of Variance	0.332790	0.252470	0.207790	0.1323300	0.0746300
Cumulative Proportion	0.332790	0.585260	0.793040	0.9253700	1.0000000

Note. Own elaboration based on ISTAT, SACE and WDI data (2024).

In Table 8, the performance of the principal components demonstrates a highly balanced and healthy distribution of total sample variance: PC_1 captures 33.28% of the total variance in the data matrix, while PC_2 captures 25.25% of the remaining variance. Together, PC_1 and PC_2 account for 58.53% of the total structural variation in the firm-destination dataset. In high-dimensional micro-data, a cumulative footprint approaching 60% across just two components is considered highly robust. In Table 9, the rotation matrix reveals exactly which underlying economic forces govern the two selected dimensions, mapping onto two distinct axes: looking across the PC_1 row values, the component is dominated almost exclusively by the firm-level topological metrics. *eigenvector* (0.6859) and *out_strength* (0.6610) exhibit massive, highly positive loadings, with a secondary positive contribution from *out_degree* (0.2519). Conversely, macro-gravity features drop to near-zero (-0.1030 for geographic distance and -0.1365 for GDP). Thus, PC_1 operates strictly as a measure of a firm's operational size and prestige inside regional networks. On the second component, the network metrics fall into statistical irrelevance. Instead, *Distance_km* (0.7018) and *GDP_Eur* (0.6915) take total control of the eigenvector rotation. Because distance and economic size load positively together on this axis, PC_2 acts as a composite index of geographical outreach, cleanly capturing destination gravity features.

Table 9. Loadings Matrix Resulting From Principal Component Analysis

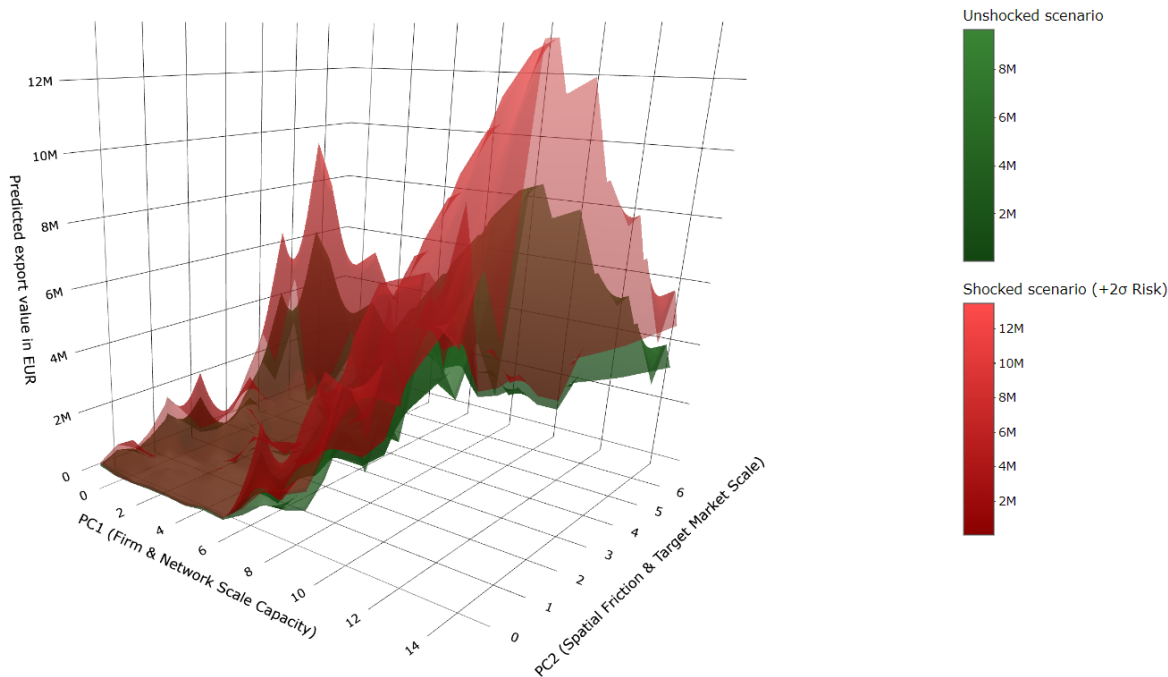
	PC1	PC2	PC3	PC4	PC5
<i>Distance_km</i>	-0.1030097	0.70179382	-0.29077282	-0.64203771	0.01063884
<i>GDP_Eur</i>	-0.1365197	0.69155289	0.29565378	0.64317536	-0.04506967
<i>out_degree</i>	0.2519523	0.04169859	-0.86533412	0.39415806	-0.17497869
<i>out_strength</i>	0.6609509	0.09599811	0.27254029	-0.13579562	-0.67912411
<i>eigenvector</i>	0.6858682	0.13522439	0.07041777	0.01766403	0.71135676

Note. Own elaboration based on ISTAT, SACE and WDI data (2024).

Crucially, the destination-side Institutional Risk Index was explicitly omitted from the exploratory Principal Component Analysis used to construct the horizontal visual dimensions. Because the stress test framework treats institutional risk as an independent, simulated treatment vector, embedding it within the PCA matrix would distort the underlying eigenvector rotation when shifting from baseline to shock configurations. Excluding the risk variable ensures that the horizontal plane functions as a completely

invariant, static structural landscape. This choice holds spatial friction and firm network assets rigid across both states, fulfilling the strict *ceteris paribus* requirements of the macro-simulation. By analyzing the continuous vertical gap between the unshocked green surface and the post-shock red surface across this fixed structural map, we can mathematically unpack the asymmetric boundaries of trade resilience.

Figure 31. Systemic Stress Test 3D Plot: Destination-Side Institutional Risk Shock



Note. Own elaboration based on ISTAT, SACE and WDI data (2024).

The 3D surface plot presented in Figure 31 provides an advanced micro econometric mapping of the systemic vulnerability within Veneto's strategic export sectors. By projecting the conditional expectations derived from the high-dimensional PPML gravity framework onto an invariant horizontal coordinate space, the visualization maps out the exact layout of trade contraction under severe institutional pressure. The horizontal dimensions are governed by two orthogonal eigenvectors: PC_1 (Firm & Network Scale Capacity) on the left axis and PC_2 (Spatial Friction & Target Market Scale) on the right axis. The vertical dimension measures the Predicted Export Value in EUR. The chart displays two distinct topological layers: the green envelope represents the unshocked baseline scenario, while the red deformed layer captures the shocked scenario under a $+2\sigma$ Institutional Risk Index increase. The most striking feature of this multi-dimensional matrix is its non-linear, ridge-like structure. Rather than showing a flat or uniform

gradient, the industrial landscape is characterized by a prominent "vulnerability ridge" that peaks sharply as network capacity and market scale expand. The green surface traces the normal operational bounds of the Veneto export performance. At low values of PC_1 and PC_2 (the bottom-left corner), the surface lies flat and close to the zero-trade floor. This section represents the fragmented SME perimeter—low-capacity, isolated firms shipping to smaller, localized markets. As we advance along PC_1 (increasing firm scale and network prestige) and PC_2 (increasing target market scale), the surface experiences an exponential surge, climbing toward peak export intensities exceeding 12 million Euros per observation cell¹³. This ridge isolates Veneto's high-capacity, hyper-connected "superstar" network systems integrators who dominate massive global hubs. When an extreme $+2\sigma$ institutional risk shock is introduced, the model applies a multiplicative penalty via the exponential link function:

$$\widehat{Y}_{\text{Shock}} = \widehat{Y}_{\text{Baseline}} \times \exp(2\beta_{\text{Risk}}). \quad (11)$$

Because the empirical PPML coefficient for institutional risk is positive ($\beta \approx 0.166$), multiplying it by a positive shock expands the exponent. Indeed, regional "superstar exporters" actively exploit high-risk/high-yield developing economies due to their capacity to navigate sunk costs. Every structural point on the grid moves perfectly vertically toward the baseline floor. By analyzing the vertical gap between the green and red envelopes, we can unpack the structural resilience margins of the firms operating in the strategic sectors. Because the vertical axis is mapped out in raw Euro levels rather than a logarithmic scale, the plot vividly highlights where the real-world economic stakes are concentrated. The vertical gap between the two surfaces is largest at the peak of the ridge (high PC_1 and high PC_2). This proves that in absolute terms, Veneto's hyper-connected superstar firms face the largest absolute volume of capital at risk. A macroeconomic collapse in high-yield, complex foreign markets directly targets the region's core value-generating hubs, threatening billions in top-tier manufacturing trade. Conversely, looking at the low-end spectrum of the horizontal plane (low PC_1 , low PC_2), the absolute vertical gap looks minimal because both surfaces are already close to the floor. However, in relative terms, this compression is highly dangerous. Because these

¹³ grid_resolution = 20 which means that a 20-step by 20-step coordinate floor, consisting of exactly 400 intersection points, is applied. It defines the density and granularity of the math grid used to plot the data.

marginal operators work within thin transaction margins and possess lower network prestige, even a minor vertical drop can push them below critical profit thresholds, forcing immediate market exit. The irregular, fluctuating nature of the red surface as it drops highlights the non-linear interaction between the network covariates. The ridge does not collapse uniformly; instead, certain pocket regions along the network axis (PC_1) show steep vertical drops, while areas with higher network centrality hold their shape better. This visual evidence confirms that robust firm-level network architecture acts as an organic buffer, helping to protect firms from the worst impacts of destination-side institutional shocks.

CONCLUSION

This thesis has developed a comprehensive, high-resolution micro econometric and algorithmic evaluation of the export structural determinants governing the internationalization, qualitative positioning, and systemic resilience of Veneto's manufacturing base. By systematically shifting the analytical lens from traditional macro-level aggregations to enterprise-level micro-data, this work has successfully mapped out the dense networks, pricing choices, and vulnerability patterns that define Northeast Italy's export engine. The research pipeline moved through distinct methodological and empirical phases combining parametric Linear Mixed Models (LMM) to isolate structural coefficient behavior with non-parametric XGBoost algorithms designed to optimize rank-based splits and capture complex non-linear interactions. Traced the geographic distribution and network positioning of Veneto's export base and highlighted the stark contrast between a hyper-connected vanguard of specialized firms and a fragmented fringe of smaller suppliers tied to localized ecosystems. Built the baseline regressions separating the drivers of absolute export value (trade mass) from the drivers of export unit value (acting as a proxy for product quality and qualitative pricing power). Used machine learning to test the predictive boundaries of these structural metrics. Implementing an optimized, regularized XGBoost configuration validated via K -fold cross-validation confirmed that firm-level network capital and destination macro-traits contain highly robust predictive signals for global pricing power. Focused directly on specialized sectors (such as ATECO 28), using quantile regression models to investigate the severe non-linearities across different firm size strata, confirming that the impact of network capital varies significantly between smaller suppliers and top-tier systems integrators. The analysis culminated in a high-dimensional PPML gravity framework that controlled for unobserved cluster heterogeneity via interacted fixed effects ($provinci^{ateco_prefix}$) and multi-way clustering, evaluating systemic vulnerability through a simulated $+2\sigma$ macroeconomic risk shock. The empirical work demonstrates that export value is determined by a complex interaction of firm-level capacity, network positioning, and destination gravity multipliers.

Structurally, traditional macro-gravity forces retain significant control over absolute trade mass; a destination market's economic acts as the primary pull factor, while

geographical distance imposes a highly stable transport penalty across all specifications. Topologically, a firm's individual network architecture acts as a vital volume multiplier. A broad extensive margin (the number of foreign countries reached) is an absolute prerequisite for scale accumulation, but the truest driver of long-term export value is firm's network prestige. Crucially, the impact of network prestige only emerges once high-dimensional fixed effects (*provinci^ateco_prefix*) sweep out the unobserved background advantages of specialized industrial districts. This confirms that individual network capital, rather than mere geographic clustering, dictates modern export success. Institutionally, country-level risk indicators display a positive semi-elasticity, revealing that Veneto's strategic manufacturing sectors actively exploit high-risk/high-yield developing economies, where localized structural gaps generate heavy demand for direct Veneto's capital goods imports. The integration of econometric modelling and machine learning algorithms confirms that a firm's network topology is a highly robust predictor of its foreign pricing power (measured via export unit values). The regularized XGBoost engine demonstrated high predictive accuracy under strict cross-validation, selecting network attributes as dominant features in predicting margin escalation. However, the quantile regression models reveal that this predictive relationship is highly non-linear and asymmetric across the performance distribution. Network position does not benefit all enterprises equally: for the marginal SME in the lower quantiles ($\tau = 0.10$), central positioning in the network offers little or no price leverage, as these firms are frequently trapped in highly substitutable subcontracting roles. Conversely, for the superstar vanguard in the upper quantiles ($\tau = 0.90$), network prestige acts as an exceptional premium extractor. These top-tier systems integrators use their central network positioning to dictate terms, pass through costs, and capture significant qualitative pricing premiums in foreign markets.

The classic Alchian-Allen effect—often referred to as "shipping the good apples out"—predicts that adding a fixed transaction cost (such as transport friction or trade barriers) lowers the relative price of high-quality goods, prompting firms to shift their export mix toward premium, high-unit-value products when serving distant or difficult destinations. The empirical deep dive into the selected strategic sectors demonstrates that destination-side institutional risk acts as a highly non-linear, destabilizing transaction friction that severely distorts this quality-sorting mechanism. Rather than encouraging a smooth

upward shift toward premium sorting across all operators, heightened institutional risk introduces a fascinating empirical paradox for the Veneto manufacturing sector, exposed by the clashing results of our parametric and non-parametric modelling architectures: the most challenging and methodologically complex result of this thesis emerges when contrasting the determinants of product quality against absolute trade mass. In the Linear Mixed Model (LMM) with firm and destination country random effects—where the response variable is the export unit value—the marginal effect of geodetic distance is positive, whereas the marginal effect of institutional risk is negative. The positive distance coefficient perfectly satisfies the classic Alchian-Allen hypothesis: as spatial friction increases, Veneto's manufacturers systematically substitute lower-grade products with premium, high-unit-value goods to absorb fixed transport costs. However, the negative coefficient for institutional risk cleanly breaks this logic. It reveals a strategic quality downgrading regime; when faced with heightened destination-side volatility (banking, corporate, sovereign, transfer, violence and expropriation risks), firms do not upgrade their quality mix. Instead, they strategically export lower-value, more substitutable commodities to volatile nations to minimize their financial exposure and cushion their margins against potential default or capital seizure. Yet, this behavior stands in sharp structural contrast to our absolute trade volume estimations. In both the high-dimensional PPML gravity framework and alternative value-based LMM configurations, the marginal effect of institutional risk on absolute export value is robustly positive. This creates a profound econometric tension: how can institutional risk simultaneously depress a firm's export unit value while expanding its absolute export revenue?

This duality implies that the Alchian-Allen sorting effect is deeply fragmented by a firm's position within the industrial topology. Veneto's strategic manufacturing sectors do not retreat from institutional risk. Rather, firms possess an advanced capabilities-driven framework to navigate the steep sunk costs of volatile jurisdictions. While the micro-level unit values drop—reflecting either a defensive compositional shift toward lower-quality goods or competitive price concessions to navigate risk frictions—the absolute level-value extraction expands. Top-tier enterprises leverage their central Network Prestige to maintain high export flows, successfully capturing market mass in complex economies where structural capital-goods deficits generate heavy demand for Veneto's medium and high technology goods. This structural polarization is clearly illustrated by the surfaces in

the 3D plot for systemic stress test. When subjected to an extreme $+2\sigma$ institutional risk shock, the continuous export surface deforms upward perfectly vertically. This layout demonstrates that the absolute macroeconomic exposure and risk-driven pricing distortions are heavily concentrated on Veneto's top-tier network hubs, highlighting the asymmetric boundaries of regional industrial resilience. For the marginal SME perimeter, high institutional risk represents an unmanageable operational friction that compresses profit margins and collapses their thin trade lines. For these lower-tier operators, the Alchian-Allen effect fails completely because the added friction functions as an outright entry barrier rather than a quality upgrade incentive. Conversely, for the hyper-connected superstar vanguard, institutional risk actively intensifies premium sorting. These elite firms leverage their deep network prestige to absorb the localized volatility, using their central positioning to layer explicit risk premiums onto their capital goods pricing. Ultimately, these findings show that the competitive edge of Veneto's industrial fabric is highly path-dependent and structurally polarized. The traditional background advantages of localized industrial districts are no longer enough on their own to guarantee global success. Instead, long-term international resilience relies heavily on an enterprise's individual ability to scale its network sophistication. For regional policymakers and industrial consortia, these insights suggest a pivot away from broad, untargeted export subsidies. Instead, resource allocation should focus on building structured international distribution networks for mid-sized operators, while engineering targeted financial cushions to shield top-tier systems integrators from absolute volume collapses during periods of severe global geopolitical volatility.

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